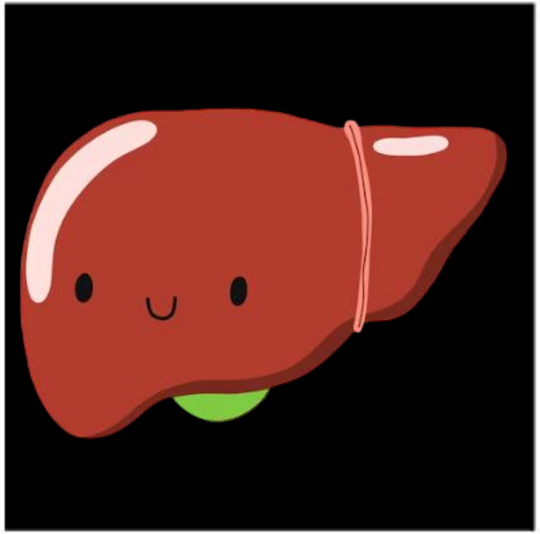
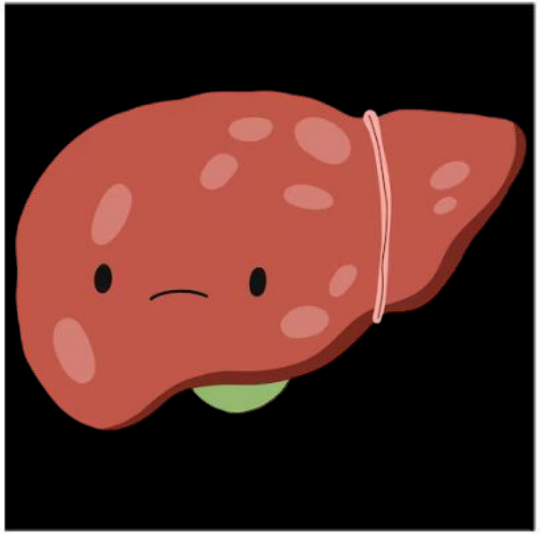
Hiperplasia Nodular Focal e HNF-like

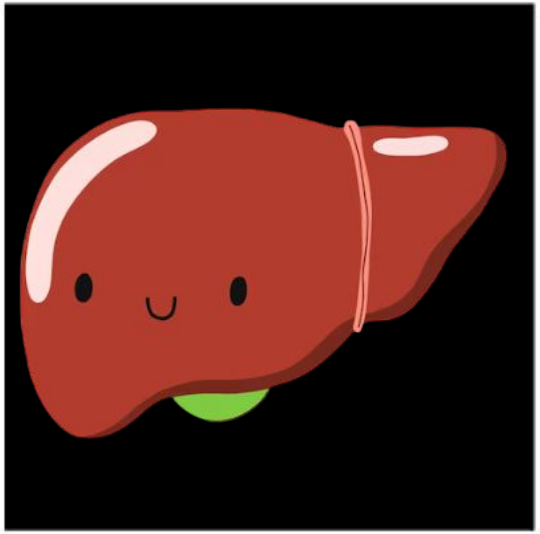
Stéfani Mariani



HNF



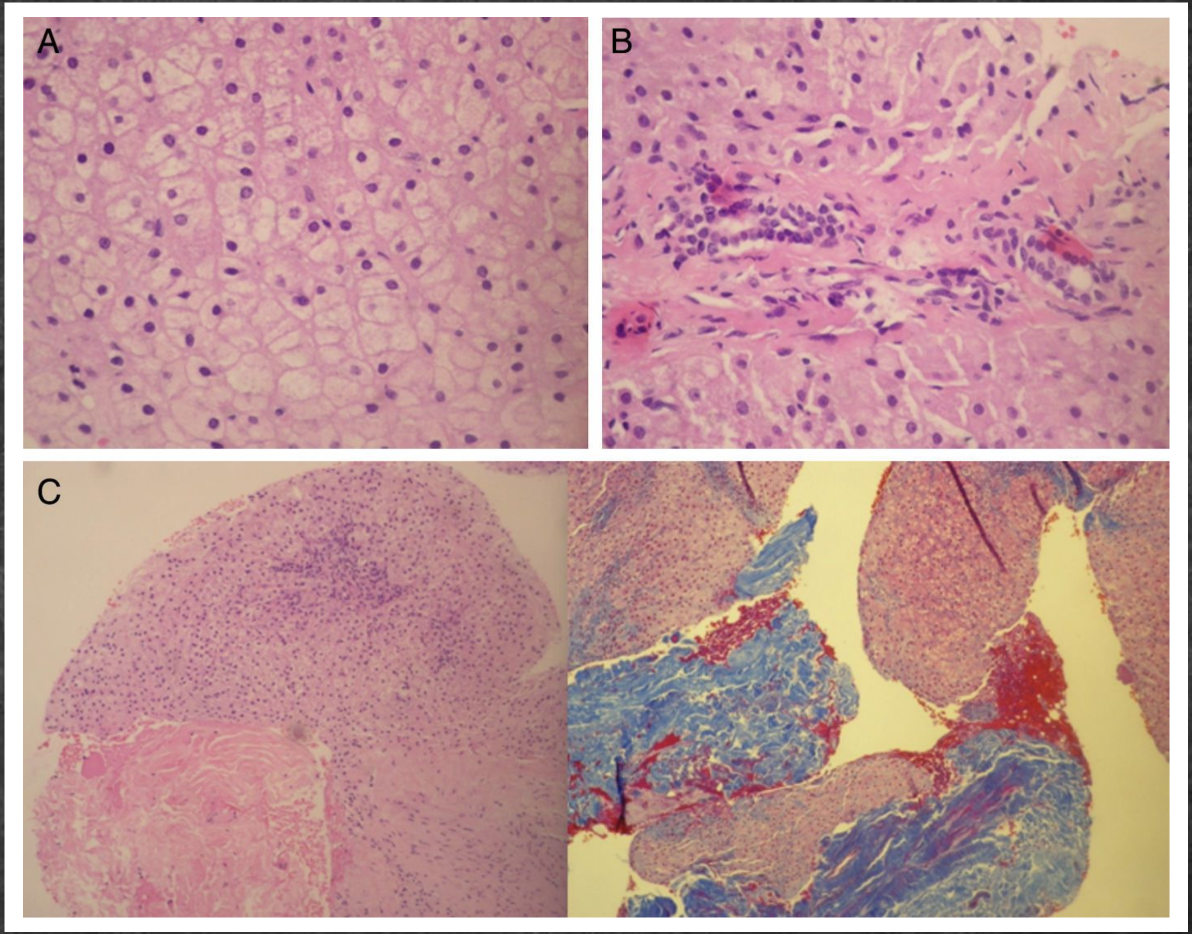
HNF-like



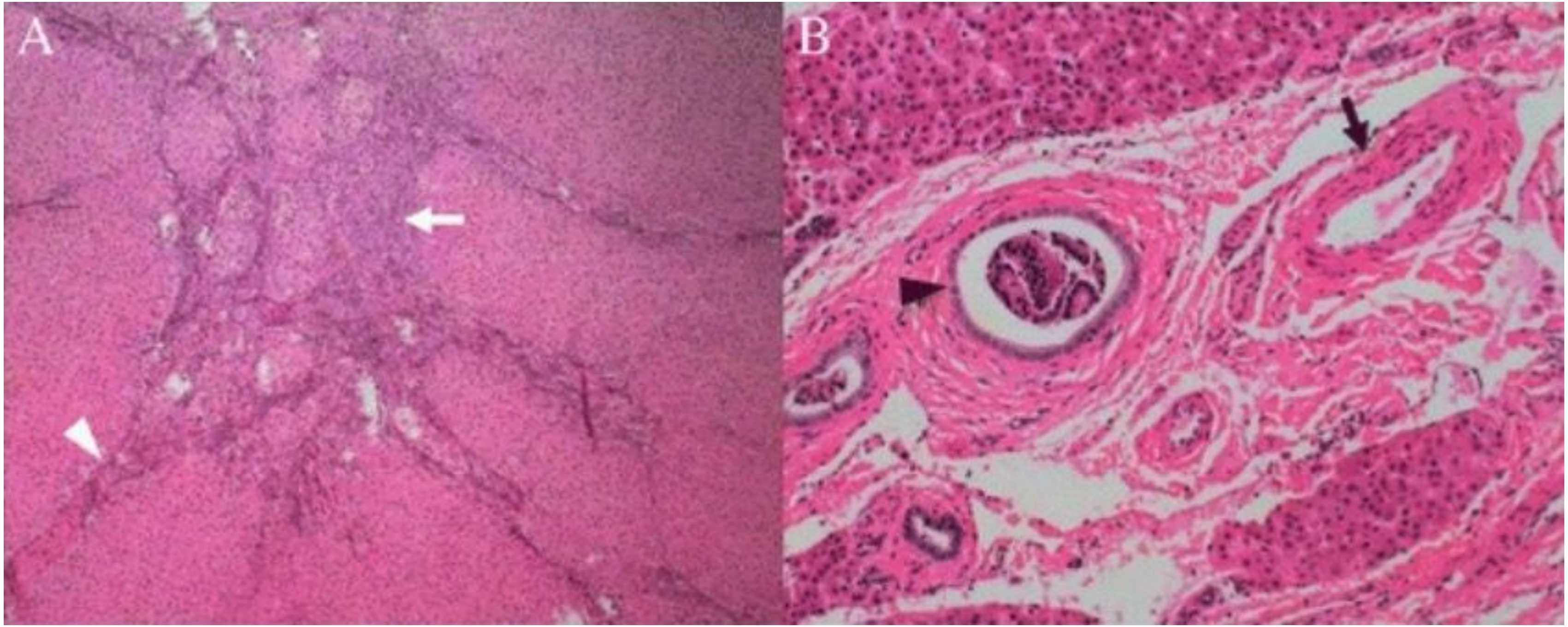
HNF

- 2ª lesão benigna mais comum do fígado;
- Mais comum em mulheres (8:1);
- 3ª a 5ª décadas de vida;
- 20% lesões múltiplas;
- Assintomática →não requer tratamento ou seguimento;
- Sintomáticas → efeito de massa → cirurgia ou embolização;

Histologia



Histologia



Características histológicas da HNF clássica: O tumor foi subdividido em nódulos por septos fibrosos (A, ponta de seta branca) com origem numa cicatriz central (A, seta branca). Artérias malformadas (B, seta preta) e proliferação ductal biliar (B, ponta de seta preta).

Imagem

- Solitária;
- Bem circunscrita;
- Não capsulada;
- Lobulada;
- <5 cm;
- Similar ao fígado → sem contraste;

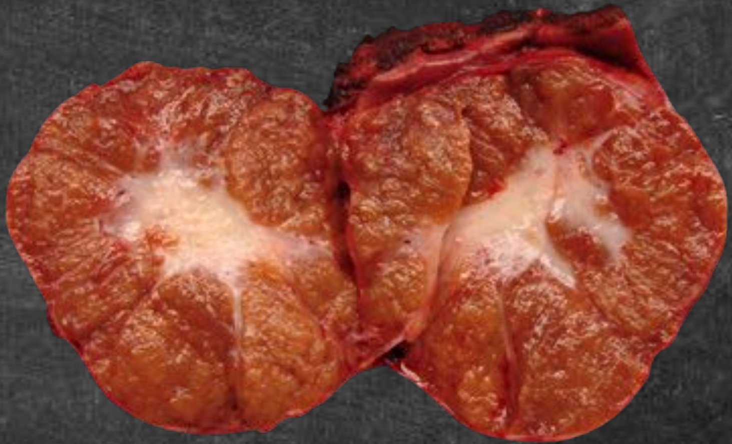


Imagem sem contraste

- USG
    - Diagnóstico acurado ~1/3 dos casos;
    - Doppler pode melhorar a visualização;
        - artéria supridora central;
        - padrão de roda de raios dos vasos ramificados nos septos radiais;
- TC
    - iso ou levemente hipodensas;
- RM
    - T1 → iso ou moderadamente hipointensas;
    - T2 → levemente hiperintensas;

Imagem com contraste

- USG
  - Hiper-realce arterial, centrífugo, realce em roda de raios ou vaso supridor excêntrico;
  - Acurácia diagnóstica de 87% - imagens contínuas com maior sensibilidade que as outras modalidades para detectar precocemente o realce em roda de raios;
- Fase portal ou tardia → HNF levemente hiperintensa ou se torna isointensa ao fígado adjacente.
- TC → 60% → cicatriz central
- RM → 80% → cicatriz central
- TC e RM com contraste extracelular → cicatriz pode ter hipo realce na fase arterial e portal e mostrar um hiper-realce mais tardio 2-5 minutos.

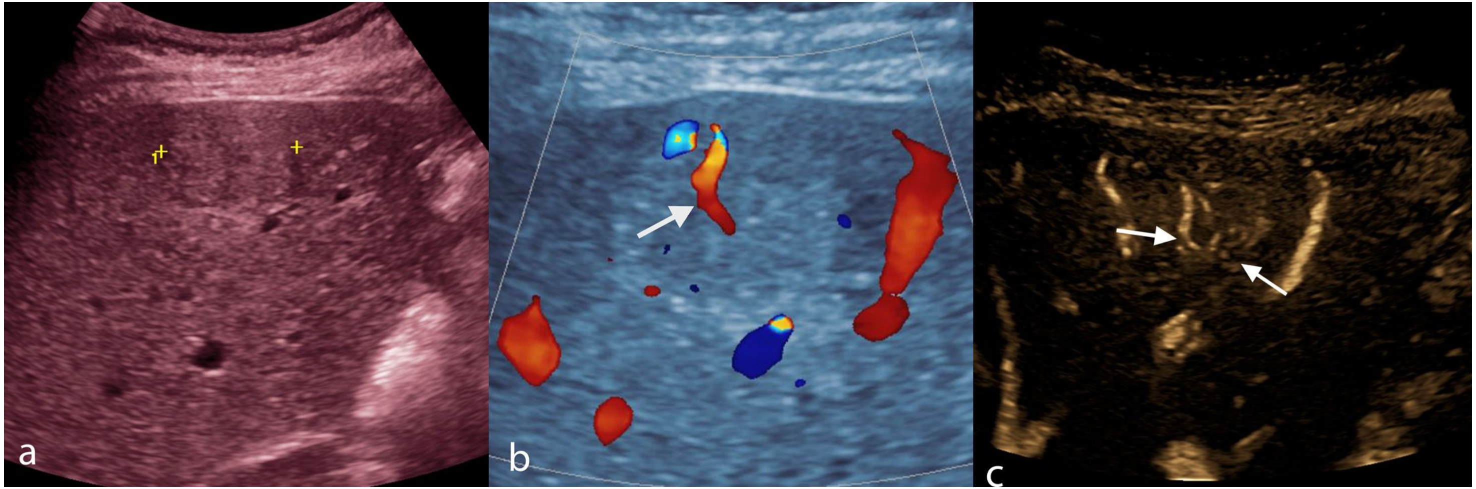


Figure E1: 46-year-old woman with incidental FNH at US. On gray scale US image with color map (a), the FNH is slightly hyperechoic to liver (calipers). Color Doppler US image (b) shows the characteristic large central feeding artery (arrow). (c) B-flow, a non-Doppler US vascular imaging technique, shows the radial vessel pattern associated with FNH (arrows).

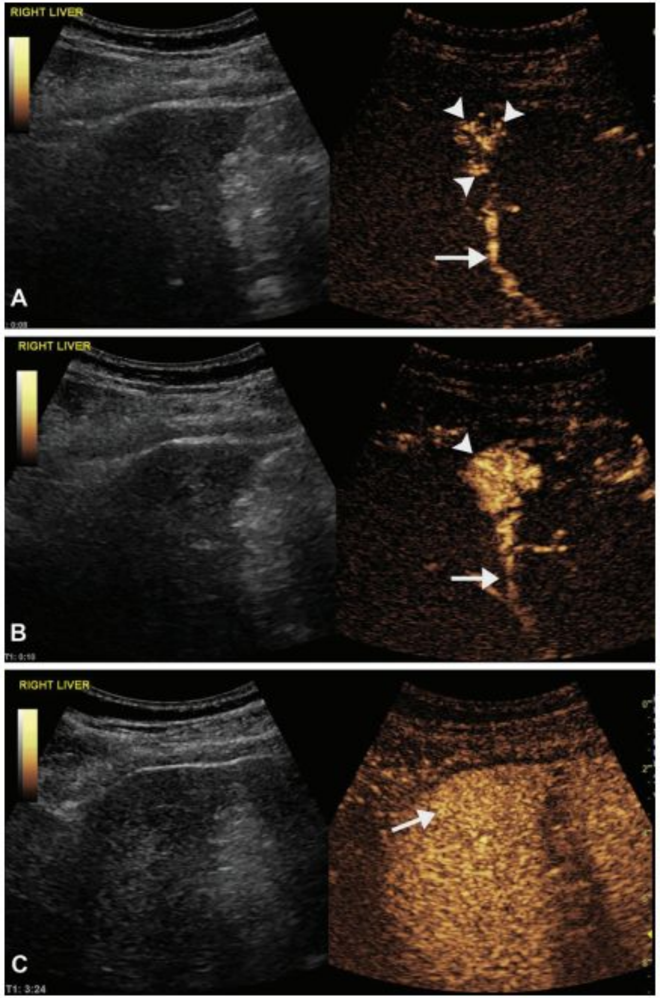


Figure 1. Focal nodular hyperplasia (FNH) in a 70-year-old woman. **(A)** Dual-screen contrast-enhanced US images at 8 seconds after administration of a microbubble contrast agent (sulfur hexafluoride lipid-type A microspheres) show the characteristic large feeding artery (arrow) and spoke-wheel enhancement pattern (arrowheads). **(B)** Dual-screen contrast-enhanced US images at 10 seconds show the hyperenhancing mass with the large feeding artery (arrow), and spoke-wheel enhancement (arrowhead) is still visible. **(C)** Dual-screen contrast-enhanced US images at 3 minutes and 24 seconds show that the lesion is nearly isoenhancing in comparison with the liver (arrow).

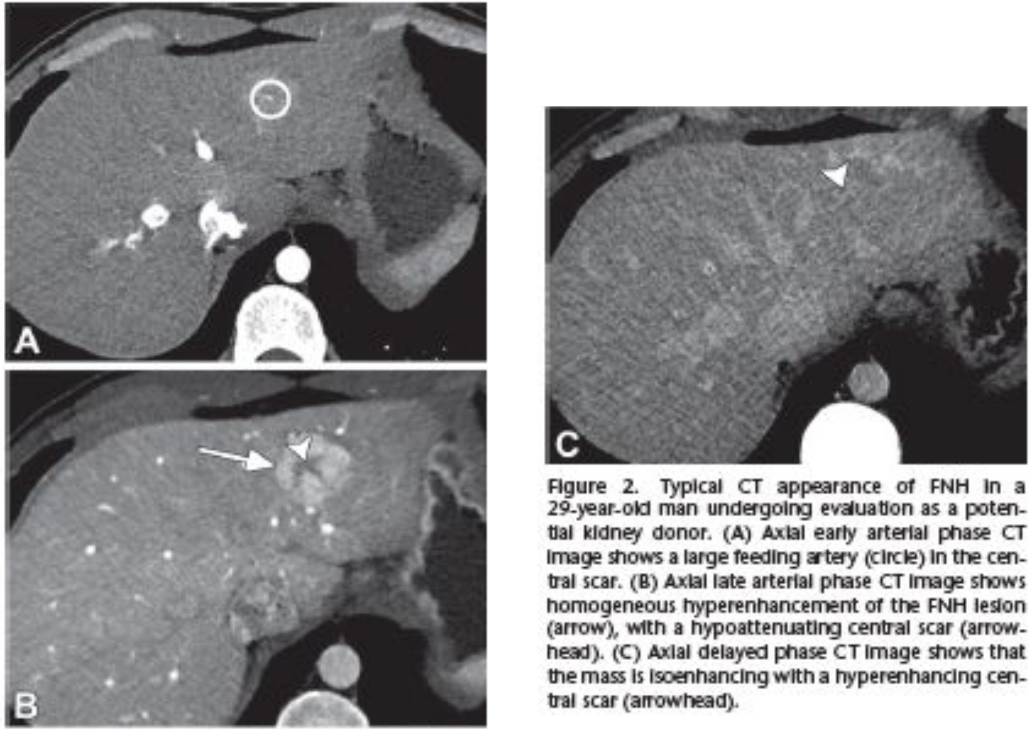


Figure 2. Typical CT appearance of FNH in a 29-year-old man undergoing evaluation as a potential kidney donor. (A) Axial early arterial phase CT image shows a large feeding artery (circle) in the central scar. (B) Axial late arterial phase CT image shows homogeneous hyperenhancement of the FNH lesion (arrow), with a hypoattenuating central scar (arrowhead). (C) Axial delayed phase CT image shows that the mass is isoenhancing with a hyperenhancing central scar (arrowhead).

RM

- Método mais sensível e específico → contraste hepato específico;
- Ponderada em difusão → levemente hiperintensa em comparação com o fígado normal;
  - Coeficientes de difusão aparentes mais elevados do que os de lesões malignas e adenomas hepatocelulares;
  - Existe uma sobreposição substancial nos coeficientes de difusão aparente, e as características na RM ponderada em difusão não são um recurso de diferenciação confiável isoladamente;
  - Cicatriz central hiperintensa em mapas de coeficiente de difusão aparente;
- Cicatriz central:
  - T1 → hipointensa;
  - T2 → hiperintensa.

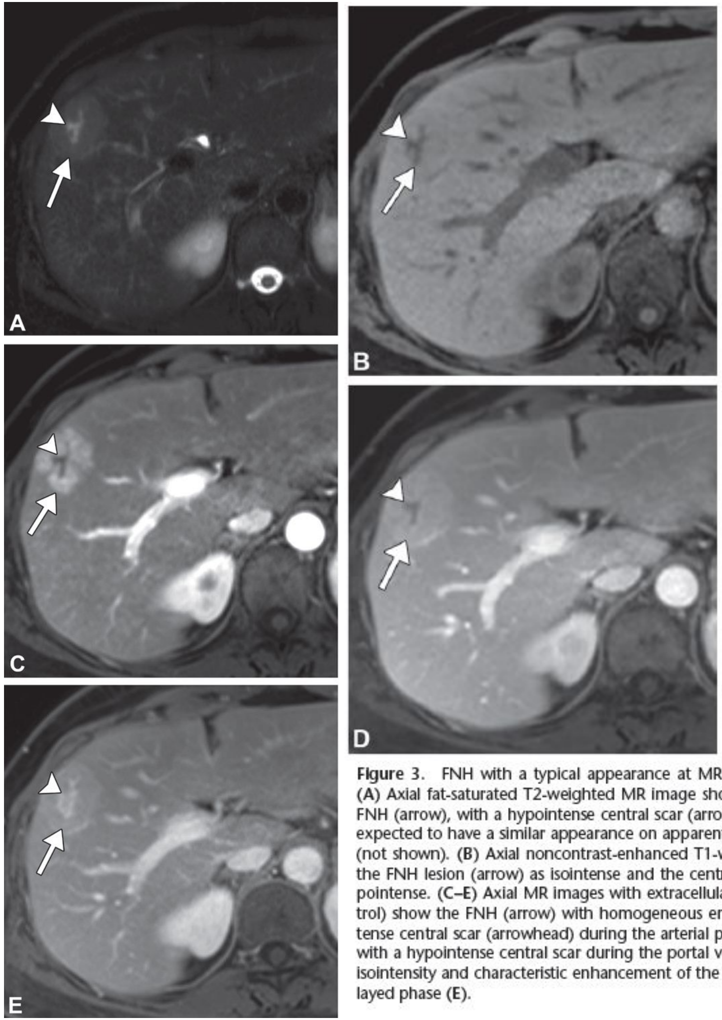
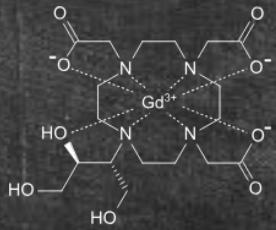
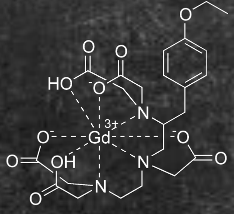


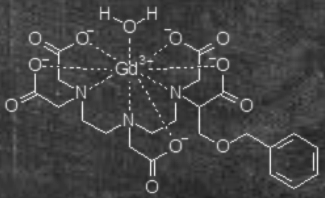
Figure 3. FNH with a typical appearance at MRI in a 46-year-old woman. **(A)** Axial fat-saturated T2-weighted MR image shows a slightly hyperintense FNH (arrow), with a hypointense central scar (arrowhead). The central scar is expected to have a similar appearance on apparent diffusion coefficient maps (not shown). **(B)** Axial noncontrast-enhanced T1-weighted MR image shows the FNH lesion (arrow) as isointense and the central scar (arrowhead) as hypointense. **(C–E)** Axial MR images with extracellular contrast agent (gadobutrol) show the FNH (arrow) with homogeneous enhancement and a hypointense central scar (arrowhead) during the arterial phase (**C**); near isointensity, with a hypointense central scar during the portal venous phase (**D**); and near isointensity and characteristic enhancement of the central scar during the delayed phase (**E**).



Contraste Hepato Específico

- Atua como contraste extracelular durante a fase venosa → fazendo a transição para a captação ao longo do tempo;
- A captação pelos hepatócitos e excreção do contraste hepato específico na bile são facilitadas por proteínas de transporte de membrana específicas dos hepatócitos, ausentes em outros tipos de células;
- Fase hepatobiliar → 20 minutos após a administração do ácido gadoxético (Primovist ®) ou 120 minutos após a administração de gadobenato dimeglumina (MultiHance ®).





RM

- HNF → captação na fase hepatobiliar → hiperintensa;
- Hemangiomas ou metástases → não captam o contraste hepato específico e ficam hipointensas durante a fase hepatobiliar;
- Outras lesões de origem hepatocelular (CHC) → usualmente são hipointensas durante a fase hepatobiliar, mas podem reter ou ter proteínas de transporte de membrana específicas de hepatócitos supra reguladas e, portanto, podem ser isointensas ou mesmo hiperintensas (minoria);
- O padrão de realce da fase arterial e venosa portal da HNF com contraste hepato específico e contrastes extracelulares é idêntico. No entanto, a cicatriz central geralmente permanece hipointensa com Primovist, à medida que começa a transição das características extracelulares para a captação específica do hepatócito antes do período típico de realce tardio da cicatriz central

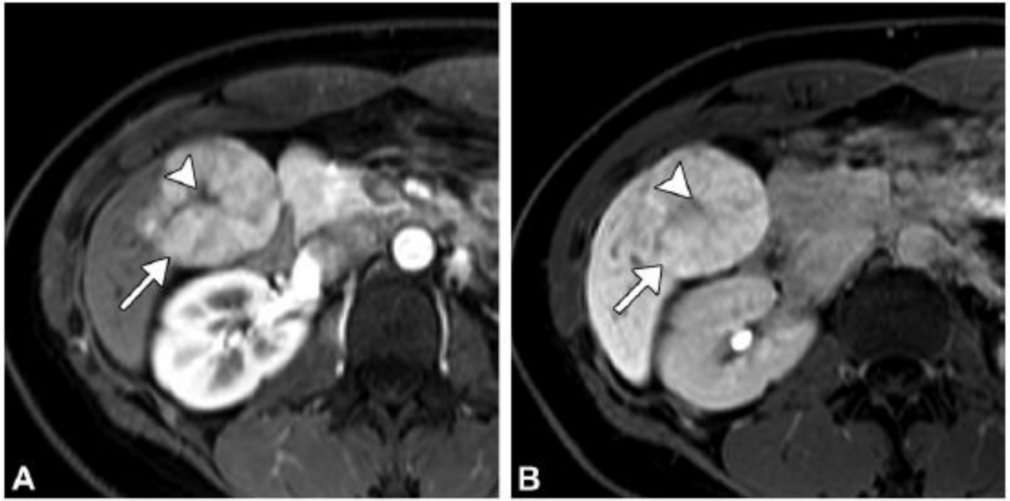


Figure 4. FNH in a 30-year-old woman, with a typical appearance at MRI. **(A)** Axial MR image shows characteristic homogeneous arterial phase enhancement (APHE) (arrow) with a hypointense central scar (arrowhead). **(B)** Axial MR image acquired with hepatocyte-specific contrast agent (HSCA) during the transitional phase shows near isointense enhancement (arrow), with a persistent hypointense central scar (arrowhead), as opposed to the delayed hyperenhancement seen with extracellular contrast agents. **(C)** Axial MR image acquired 20 minutes after administration of HSCA during the hepatobiliary phase (HBP) shows characteristic homogeneous hyperintense uptake of contrast material (arrow), with a persistent hypointense central scar (arrowhead).

RM

HNF
4 padrões

- Hiperintensidade homogênea;
- Hiperintensidade heterogênea;
- Isointensidade homogênea;
- Hipointensidade com captação em anel periférico;

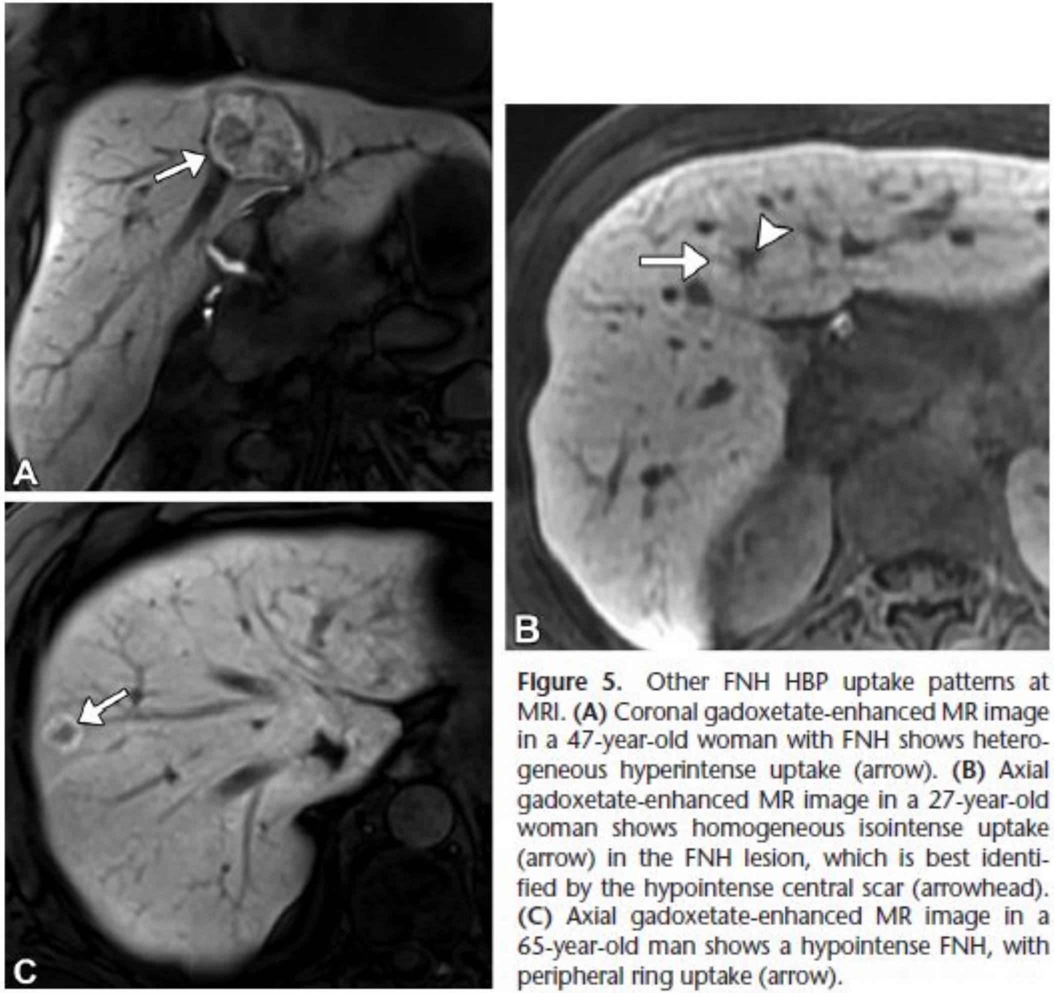


Figure 5. Other FNH HBP uptake patterns at MRI. **(A)** Coronal gadoxetate-enhanced MR image in a 47-year-old woman with FNH shows heterogeneous hyperintense uptake (arrow). **(B)** Axial gadoxetate-enhanced MR image in a 27-year-old woman shows homogeneous isointense uptake (arrow) in the FNH lesion, which is best identified by the hypointense central scar (arrowhead). **(C)** Axial gadoxetate-enhanced MR image in a 65-year-old man shows a hypointense FNH, with peripheral ring uptake (arrow).

RM

1. **Padrão em alvo** → retenção do contraste em tumores com estroma fibroso, não é captação celular → colangiocarcinoma e adenocarcinoma metastático;

2. **Realce periférico** → retenção peritumoral → captação hiperintensa ocorre em uma borda periférica de hiperplasia hepatocítica peritumoral

- Padrão em alvo → realce central de contraste com uma borda periférica hipointensa;
- Borda peritumoral hiperintensa;
- Excreção de contraste nos ductos biliares peri ou intralesionais;

Não HNF

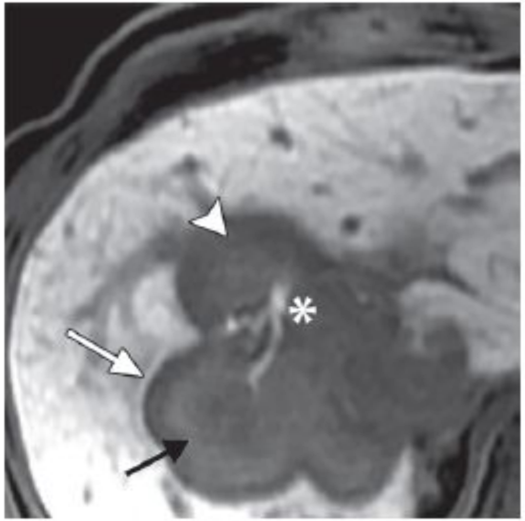


Figure 6. Intrahepatic cholangiocarcinoma in a 54-year-old man. Axial gadoxetate-enhanced HBP MR image shows the targetoid appearance of the mass, with areas of central contrast material retention (black arrow), peripheral hypointensity (arrowhead), and peritumoral retention (white arrow). These patterns should not be confused for uptake in an FNH lesion. The intrahepatic cholangiocarcinoma also shows enhancement of intralesional bile ducts (*).

Achados Atípicos de HNF

Largura > 10 cm	seguimento, considerar biópsia se diagnóstico incerto
Crescimento	seguimento, considerar biópsia se diagnóstico incerto
Esteatose	seguimento se outros achados clássicos, senão biópsia
Calcificação	seguimento se outros achados clássicos, senão biópsia
Sinal Heterogêneo T2	considerar diagnóstico alternativo e biópsia
Pouco realce arterial	considerar diagnóstico alternativo e biópsia
Hipointenso FHB	considerar diagnóstico alternativo e biópsia

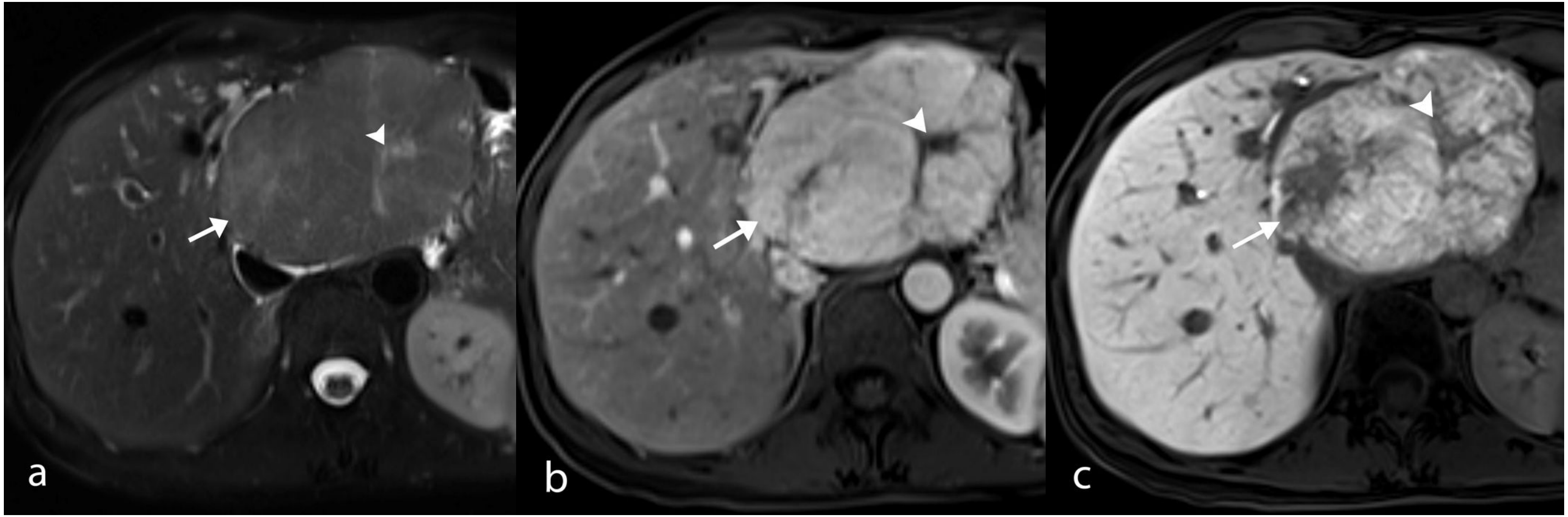


Figure E2: 49-year-old woman with an atypically large 11-cm FNH at MRI, including T2-weighted fat-saturated (a), arterial phase (b), and HBP (c) images with gadoxetate contrast agent. FNH (arrow) otherwise demonstrates typical features with a T2-hyperintense and postcontrast hypointense central scar (arrowhead), slight T2 hyperintensity (a), homogeneous APHE (b), and heterogeneous hyperintense contrast agent uptake (c). This lesion was symptomatic and surgically resected.

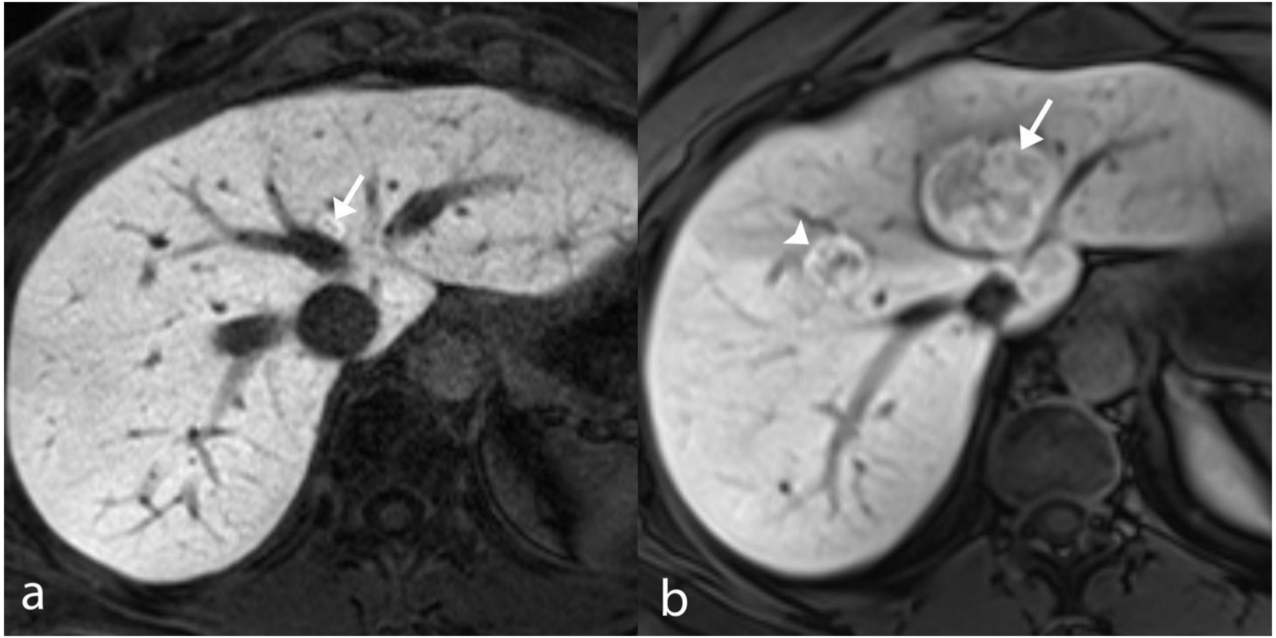


Figure E3: 43-year-old woman with growth of FNH on HBP from MRI with gadoxetate contrast agent. (a) Baseline examination demonstrates a small FNH with hyperintense uptake (arrow) in liver segment IVa. (b) MRI image at follow-up examination 4 years later shows growth of FNH in segment IVa (arrow) and a new FNH with hyperintense uptake in segment VIII (arrowhead).

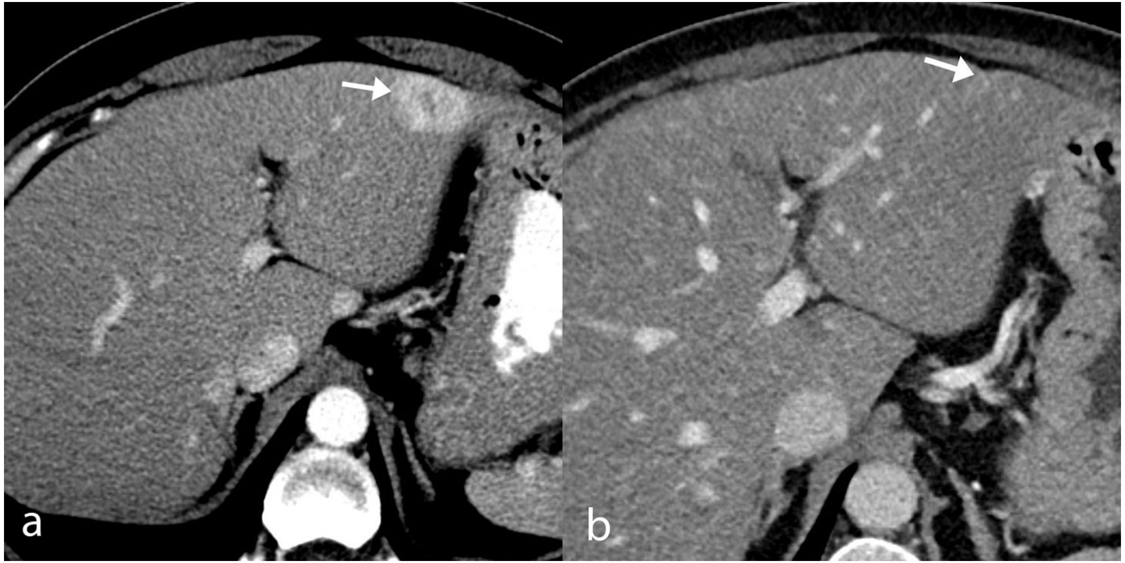


Figure E4: 44-year-old woman with involution of FNH at CT with intravenous contrast agent. (a) Baseline CT demonstrates a left hepatic lobe FNH (arrow) that involutes at follow-up 10 years later (b) with only a tiny focus of hyperenhancement and subtle capsular retraction and scarring remaining (arrow).

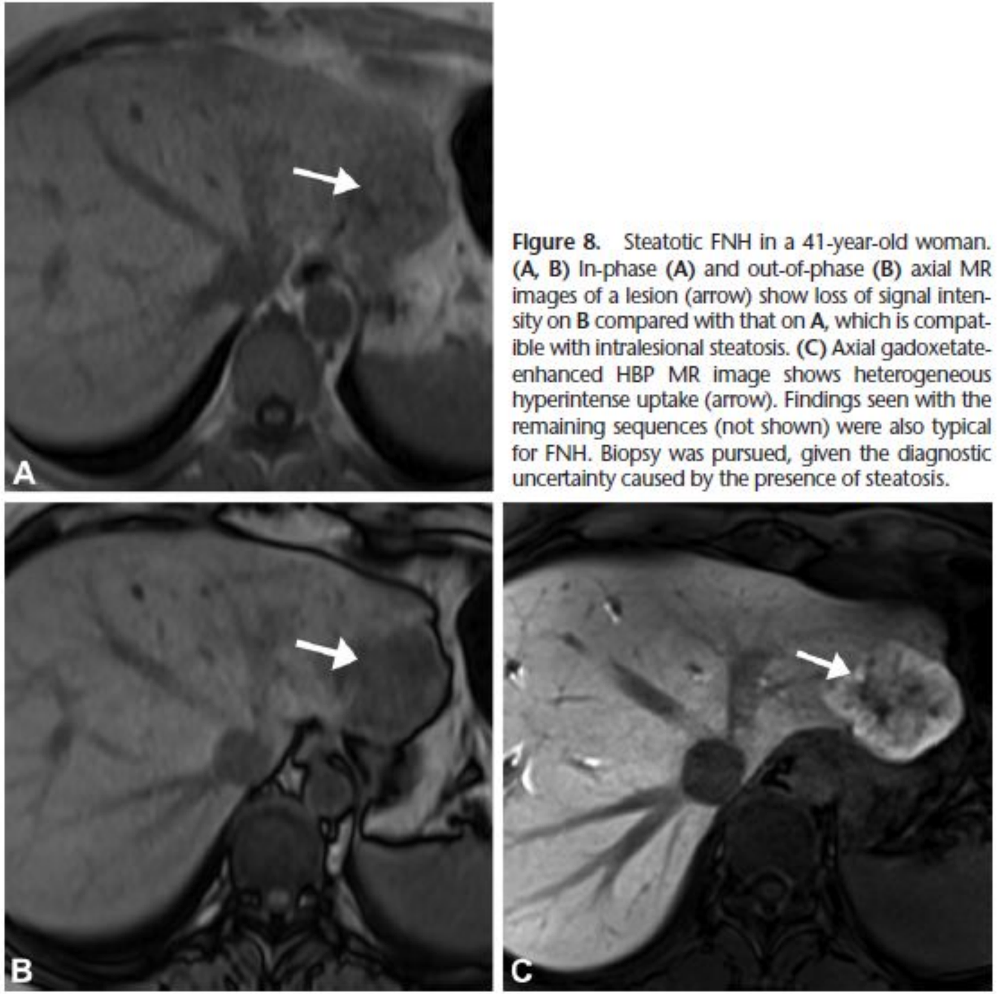


Figure 8. Steatotic FNH in a 41-year-old woman. **(A, B)** In-phase **(A)** and out-of-phase **(B)** axial MR images of a lesion (arrow) show loss of signal intensity on **B** compared with that on **A**, which is compatible with intralesional steatosis. **(C)** Axial gadoxetate-enhanced HBP MR image shows heterogeneous hyperintense uptake (arrow). Findings seen with the remaining sequences (not shown) were also typical for FNH. Biopsy was pursued, given the diagnostic uncertainty caused by the presence of steatosis.

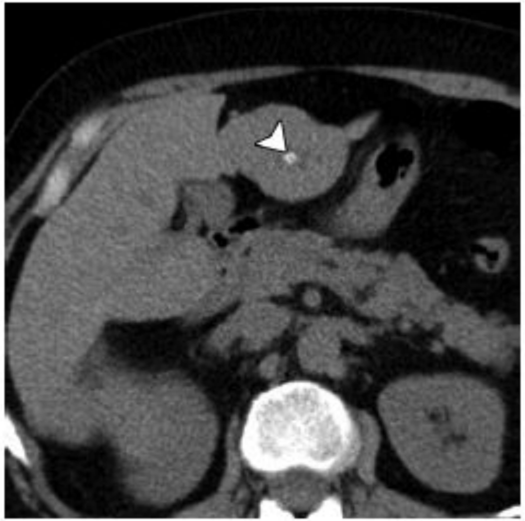


Figure 9. Left hepatic lobe FNH in a 53-year-old woman. Axial unenhanced CT image shows punctate calcification in the central scar (arrowhead). Calcification is rare in FNH and is considered an atypical feature. This lesion showed stability for 10 years and also had features at MRI that were typical of FNH (not shown).

HNF x ADENOMA

- Adenoma → risco de transformação maligna (4%) e de hemorragia (16%);
  - HNF-1α - inativado;
  - Inflamatório;
  - β-catenina ativado;
  - Não classificado;
- Diferenciação necessária em função de abordagens distintas;
- RM - com contraste hepato específico → acurácia de 97-100% nos estudos iniciais → 87-95%;
  - Adenomas → 75-90% são hipointensos durante a fase hepatobiliar → 25% dos inflamatório e 40-80% dos β-catenina ativados são iso ou hiperintensos (mimetiza HNF); 43% dos inflamatórios e 67% dos β-catenina ativados têm cicatriz central;

HNF x ADENOMA

Lesão	Gordura Intralesional	Cicatriz Central	Realce Arterial	Fase Portal ou Tardia	Fase Hepatobiliar
Hiperplasia nodular focal	Raro	Frequente	✓	Isointenso	Hiper ou isointenso
HNF1-α inativado	Frequente	X	✓	Variável	Hipointenso
Inflamatório	Variável	Variável	✓	Variável	Variável
β-catenina ativado	Raro	Variável	✓	Variável	Iso ou hiperintenso

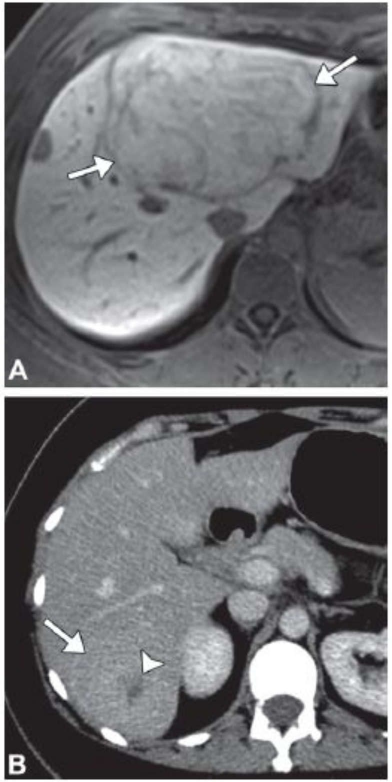


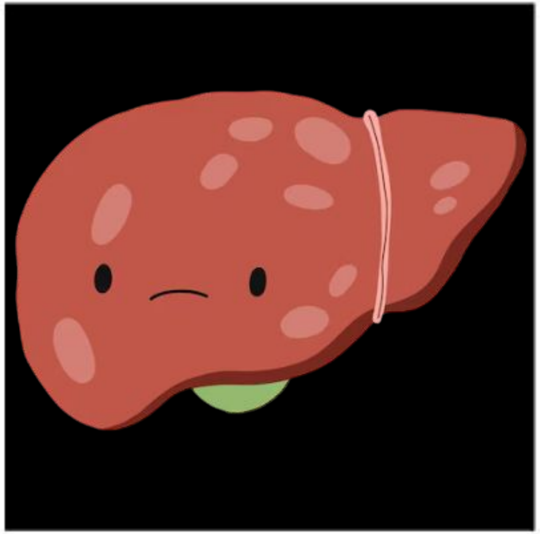
Figure 7. Hepatocellular adenoma (HCA) mimicking FNH. (**A**) Axial HBP MR image acquired at 20 minutes after administration of gadoxetate in a 21-year-old woman with a β-catenin HCA (arrows) shows isointense to slightly hyperintense uptake of contrast material, which mimics the appearance of FNH. (**B**) Axial portal venous phase CT image in a 38-year-old woman with a β-catenin HCA (arrow) shows a central scar (arrowhead) mimicking FNH.

TAKE HOME MESSAGES

- Se a lesão demonstrar achados típicos de HNF em todas as sequências, então se pode fazer um diagnóstico com mais confiança;
- Se um achado típico não estiver presente, ou tiver pelo menos um achado atípico então deve se proceder com seguimento por imagem ou biópsia tecidual, baseado no grau de suspeição de uma lesão não HNF.


10/6
I'm waiting.

- Macroscopia, microscopia e imunoistoquímica idêntica à HNF;
- Sem risco de transformação maligna;
- Condições predisponentes.



HNF-like

Histologia

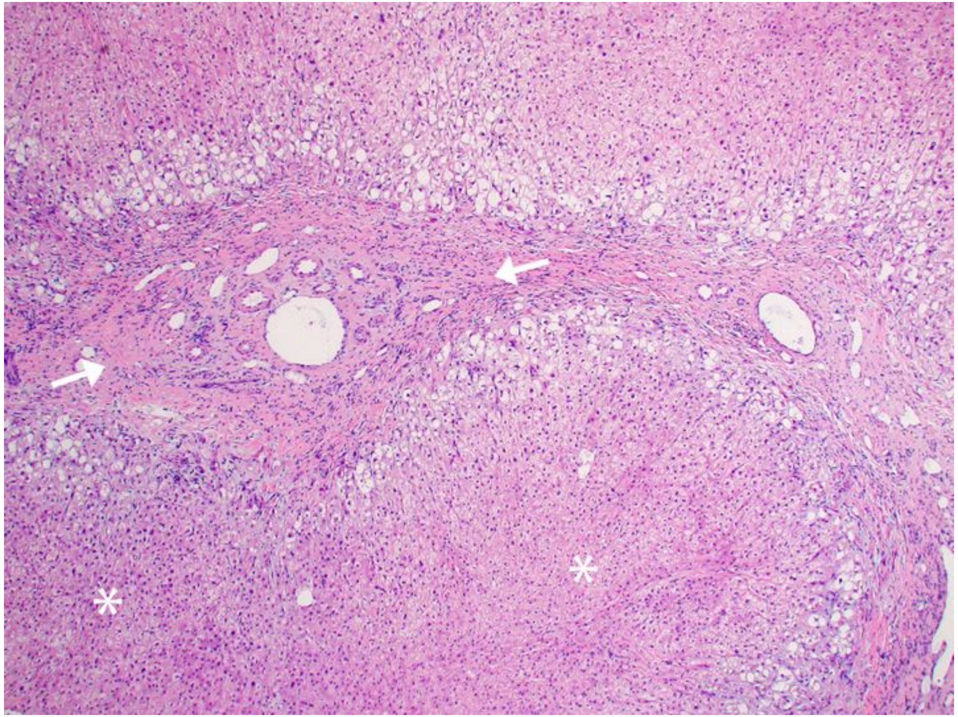
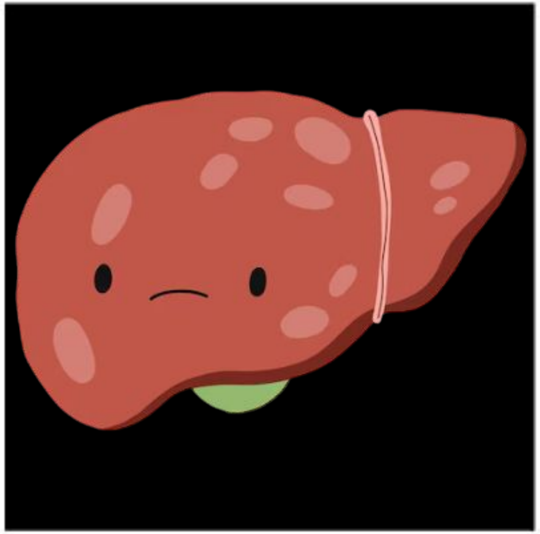
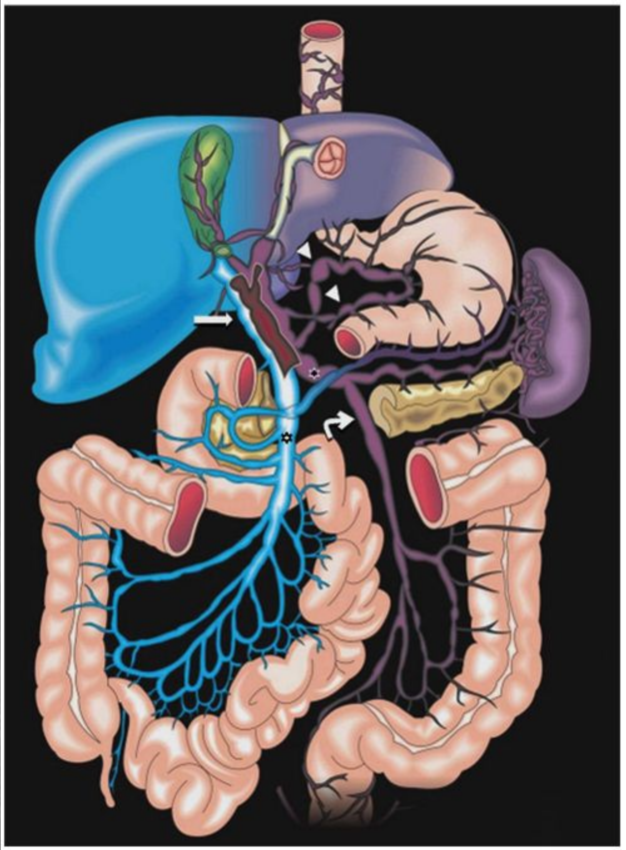


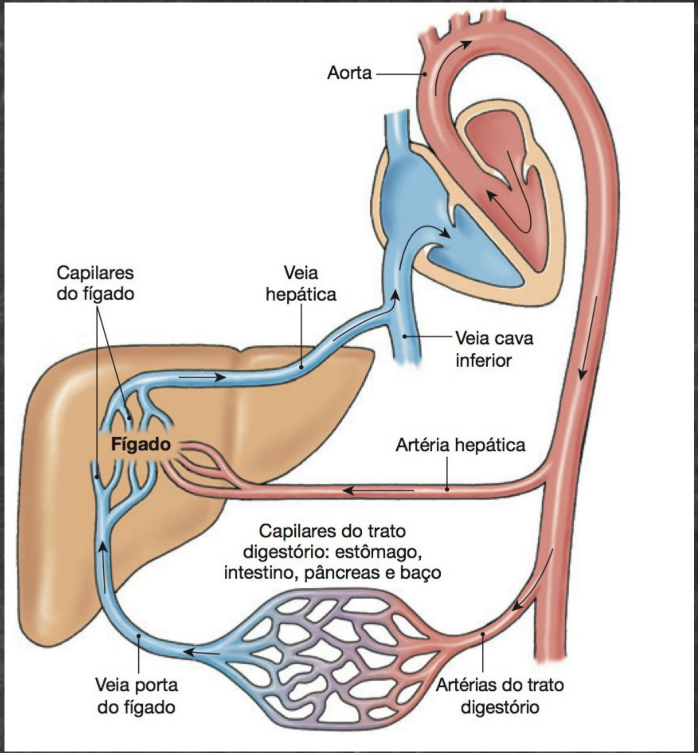
Figure E5: Low-power view of hematoxylin-eosin staining of the FNH-like lesion in Figure 14. FNH and FNH-like lesions are identical histologically, differing only in the nature of the background liver in which they occur. There is a large central scar (arrows) containing vessels of various calibers. No normal portal structures are seen. Note the nodular arrangement of hepatocytes in the lower one-half of the field (*).

Table 1: Conditions Predisposing Patients to Formation of FNH-like Lesions
Abnormalities of hepatic outflow
Budd-Chiari syndrome
Fontan-associated liver disease
Other forms of cardiogenic congestive hepatopathy
Abnormalities of hepatic inflow
Congenital absence of the portal vein
Cavernous transformation
Hereditary hemorrhagic telangiectasia
Hepatic microvascular disturbances
Cirrhosis
Nodular regenerative hyperplasia
Chemotherapy induced



HNF-like




Aorta
Capilares do fígado
Veia hepática
Veia cava inferior
Fígado
Artéria hepática
Capilares do trato digestório: estômago, intestino, pâncreas e baço
Veia porta do fígado
Artérias do trato digestório

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Abnormalities of hepatic outflow
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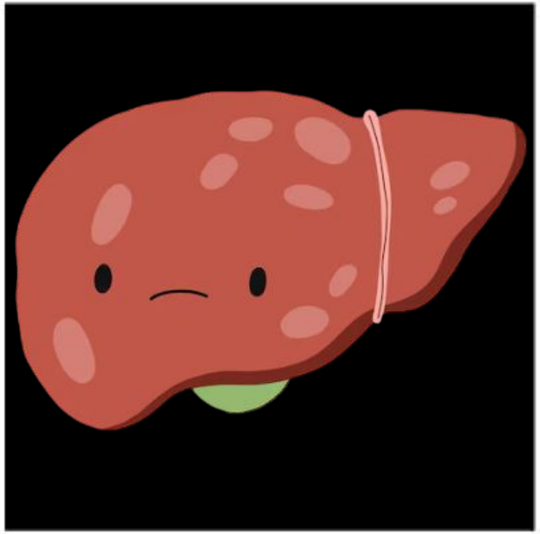
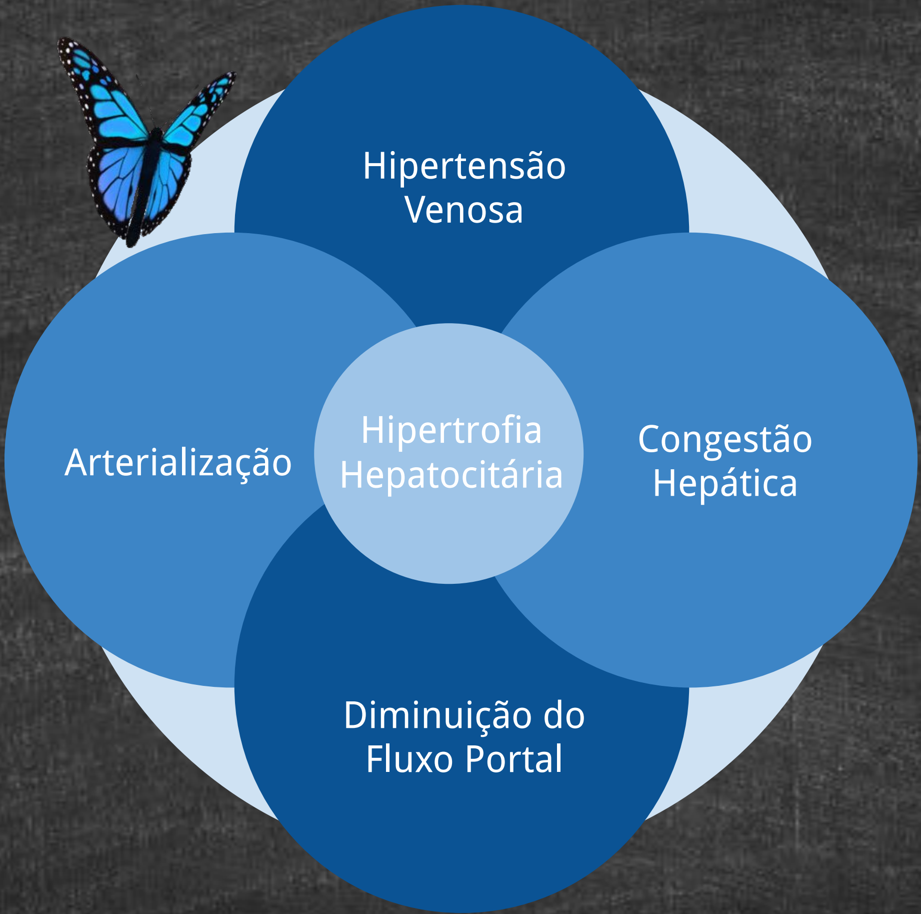


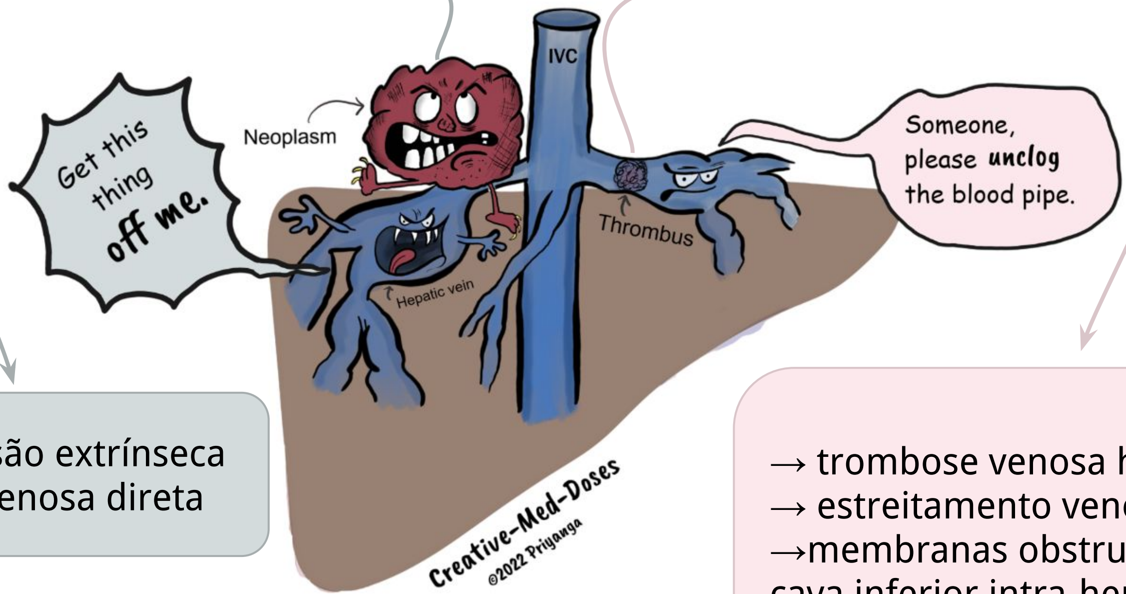
HNF-like

ALTERAÇÕES DO FLUXO DE SAÍDA



SÍNDROME DE BUDD-CHIARI

2^a



1^a

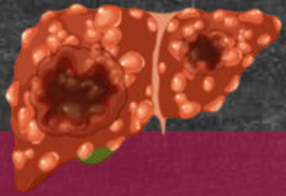
→ compressão extrínseca
→ invasão venosa direta

→ trombose venosa hepática
→ estreitamento venoso intrínseco
→membranas obstruindo a veia cava inferior intra-hepática ou veias hepáticas

SÍNDROME DE BUDD-CHIARI

- Lesões HNF-like em 1/3 dos casos;
- Costumam ser lesões múltiplas;
- RM
  - T1 → hiperintenso;
  - T2 → hipointenso;

  atípico
  - Hiper-realce na fase arterial;
  - Captação na fase HPB;
  - Subtração de gordura → melhora avaliação de captação arterial → T1 hiperintenso;
  - Sem restrição à difusão;
  - Cicatriz central;
  - 1/3 *washout.*



CHC
- T1 → iso ou hipointenso, pode ser hiper;
- Hiper-realce arterial;
- *Washout* rápido;
- HPB → geralmente hipointenso;
- T2 → variável, hiperintenso;
- DWI → alto sinal intratumoral.

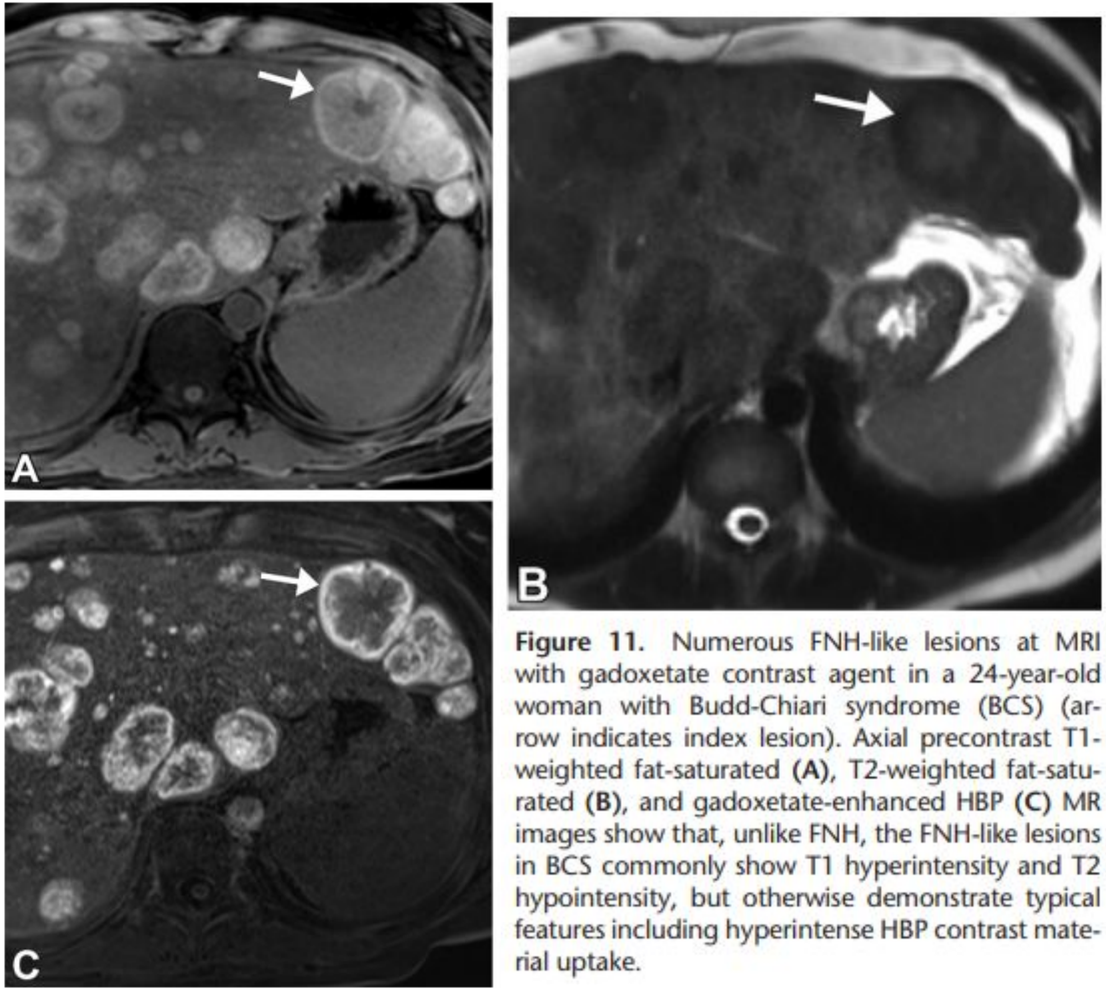
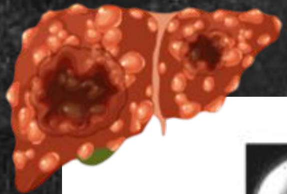


Figure 11. Numerous FNH-like lesions at MRI with gadoxetate contrast agent in a 24-year-old woman with Budd-Chiari syndrome (BCS) (arrow indicates index lesion). Axial precontrast T1-weighted fat-saturated **(A)**, T2-weighted fat-saturated **(B)**, and gadoxetate-enhanced HBP **(C)** MR images show that, unlike FNH, the FNH-like lesions in BCS commonly show T1 hyperintensity and T2 hypointensity, but otherwise demonstrate typical features including hyperintense HBP contrast material uptake.



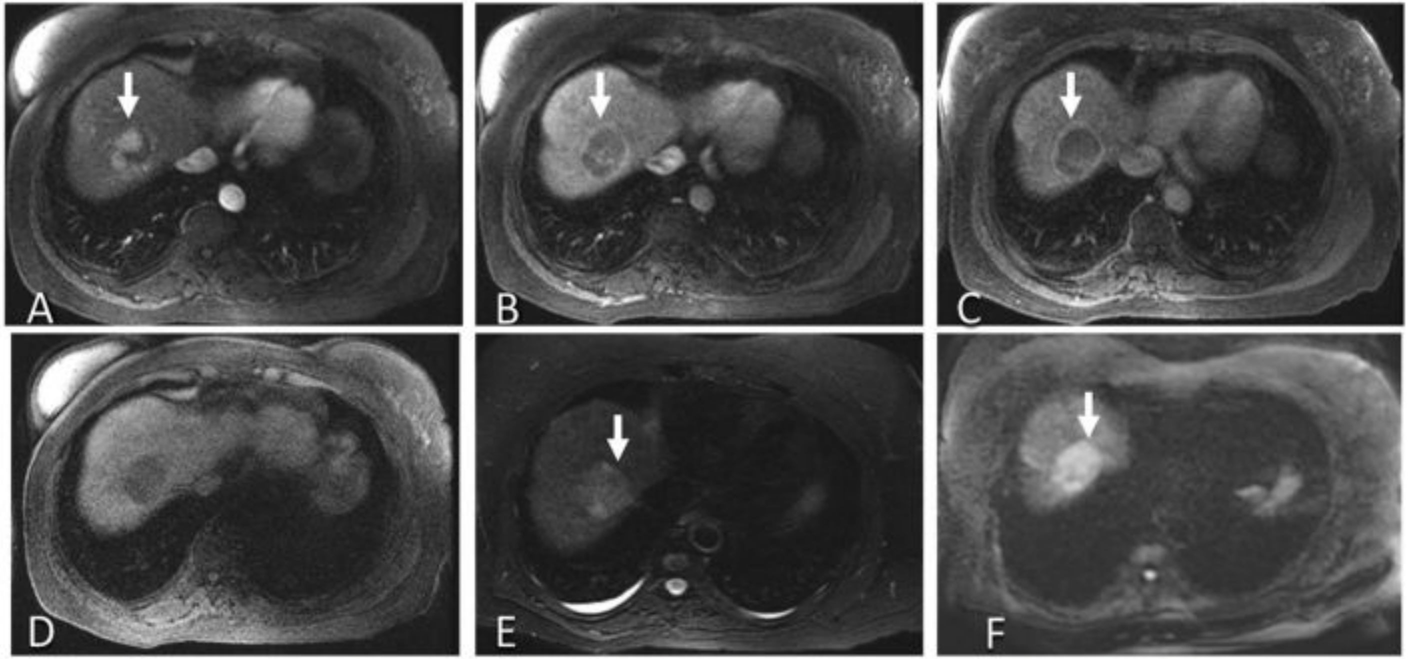
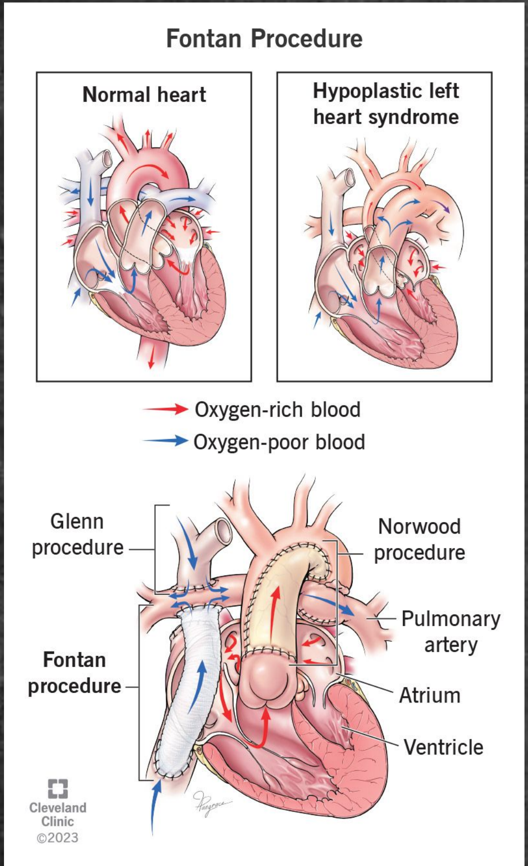
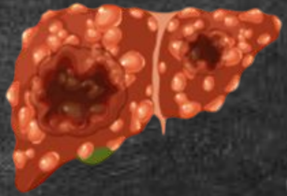


Fig. 2. Axial magnetic resonance imaging of the HCC in a 57-year-old patient with cirrhosis secondary to chronic hepatitis C infection. Arterial (A), portal venous (B), delayed (C) phase demonstrating arterial phase hyperenhancement in HCC (arrow) which washes out on portal venous and delayed phases. This mass demonstrates T1 hypointensity (D), and mild T2 hyperintensity (E) with mild restricted diffusion (F).

FONTAN-ASSOCIATED LIVER DISEASE



- HNF-like → 20-50%;
- Pequenas (~1 cm);
- Periféricas (75% fica a < 2cm da margem);
- Hiper-realce arterial;
- Múltiplas lesões;
- 7-10% *washout.*



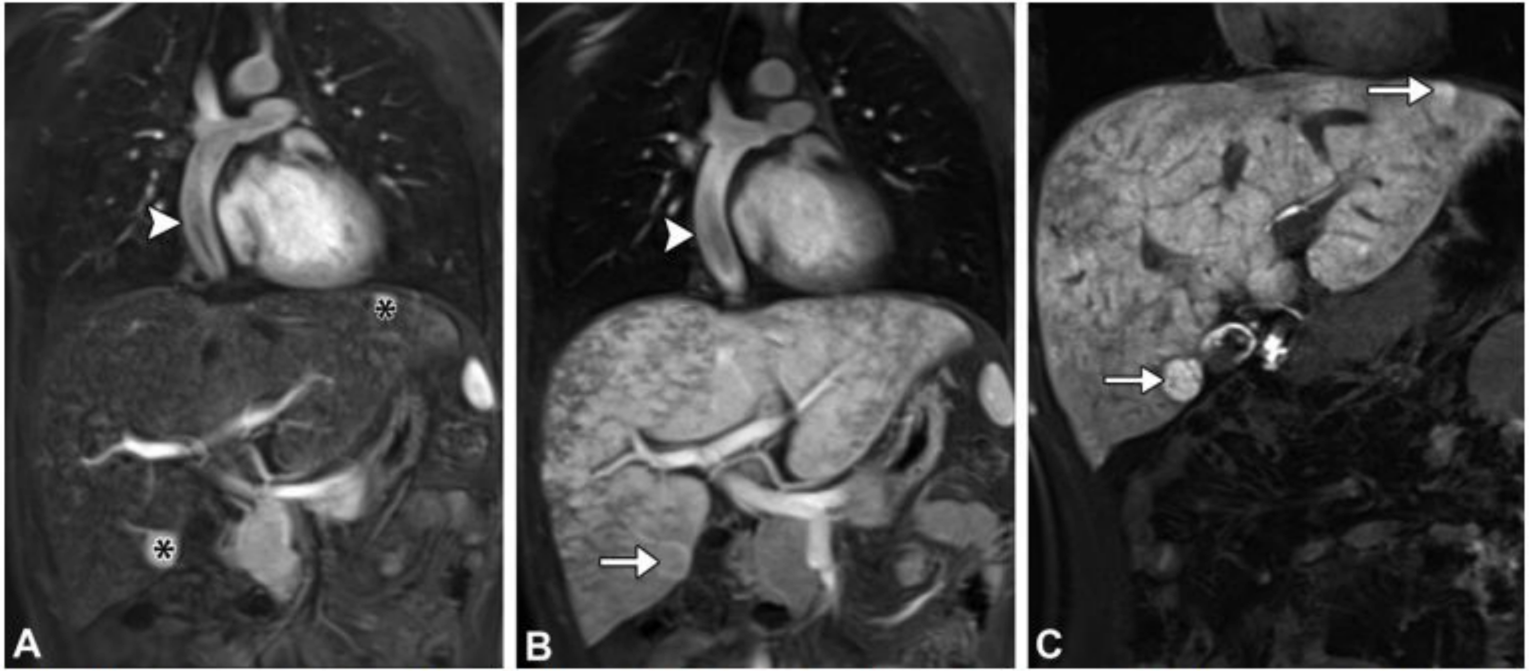
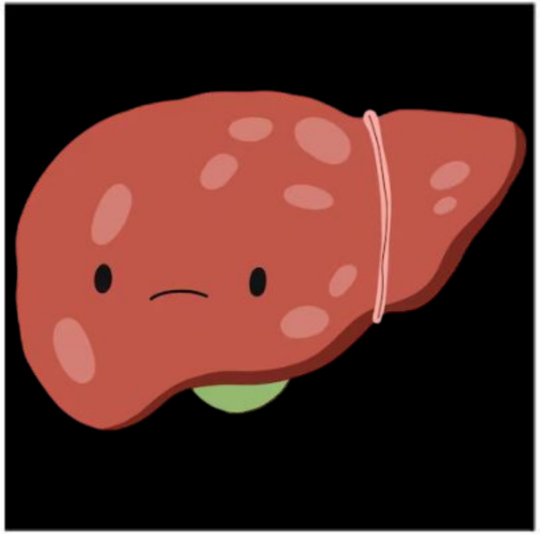


Figure 10. Images of the heart and liver in a 26-year-old man after he underwent Fontan surgery. **(A, B)** Coronal arterial phase **(A)** and portal venous phase **(B)** gadobutrol-enhanced MR images show two (of multiple) APHE liver lesions (* in **A**), one of which demonstrates washout appearance (arrow in **B**). **(C)** Subsequent coronal gadoxetate-enhanced HBP MR image shows homogeneously hyperintense uptake that is consistent with FNH-like lesions (arrows). Note the widely patent Fontan shunt (arrowhead in **A** and **B**).

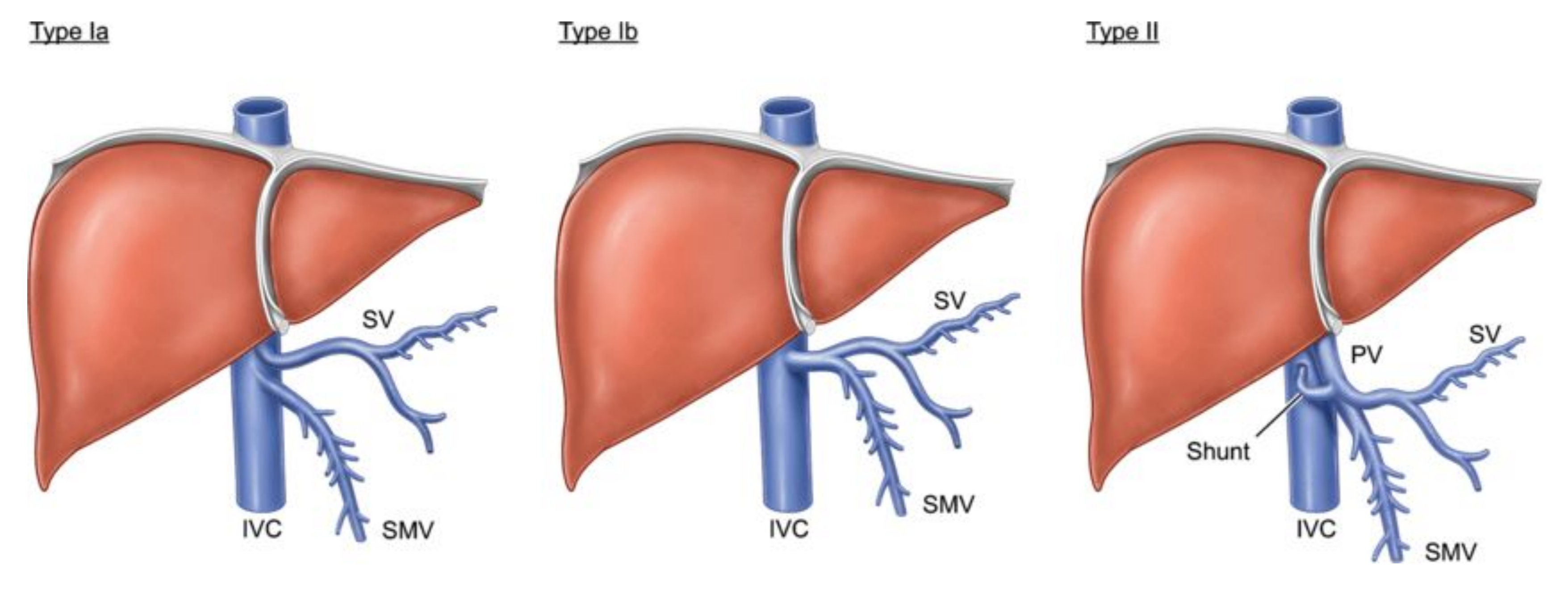
Table 1: Conditions Predisposing Patients to Formation of FNH-like Lesions

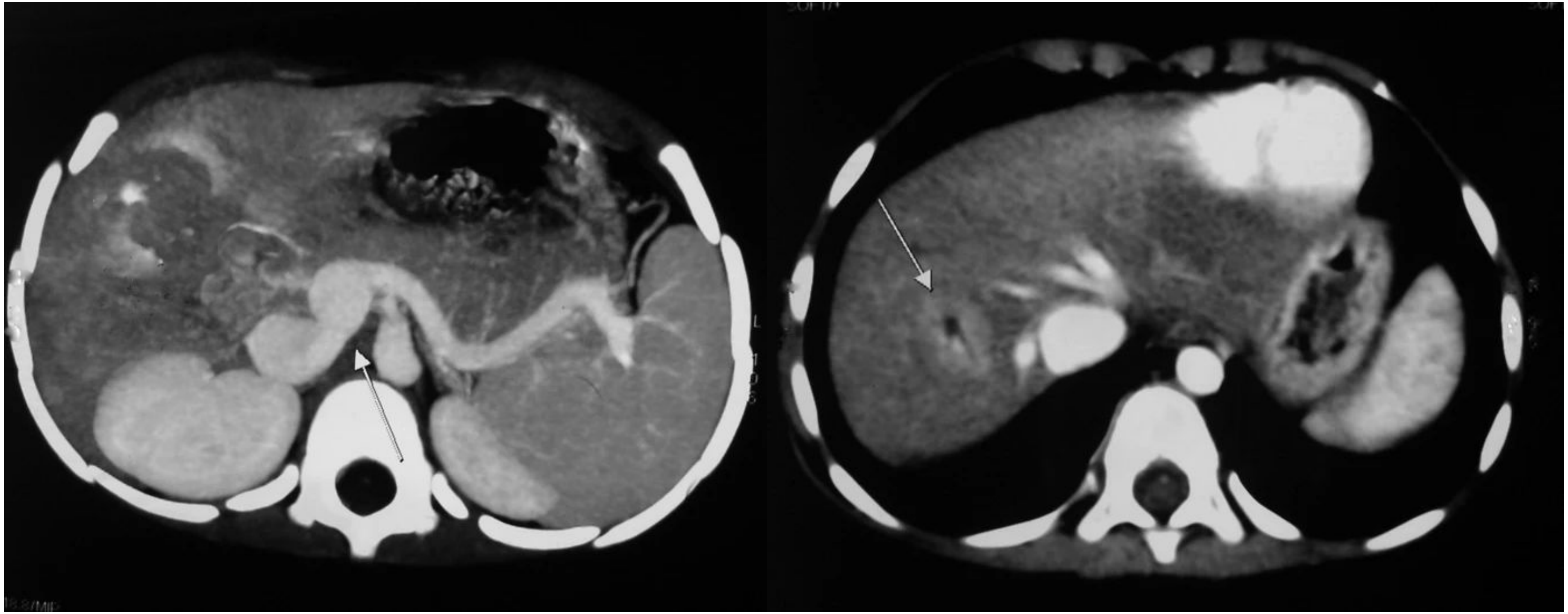
Abnormalities of hepatic outflow
Budd-Chiari syndrome
Fontan-associated liver disease
Other forms of cardiogenic congestive hepatopathy
Abnormalities of hepatic inflow
Congenital absence of the portal vein
Cavernous transformation
Hereditary hemorrhagic telangiectasia
Hepatic microvascular disturbances
Cirrhosis
Nodular regenerative hyperplasia
Chemotherapy induced



HNF-like

AUSÊNCIA CONGÊNITA DA VEIA PORTA





TRANSFORMAÇÃO CAVERNOMATOSA

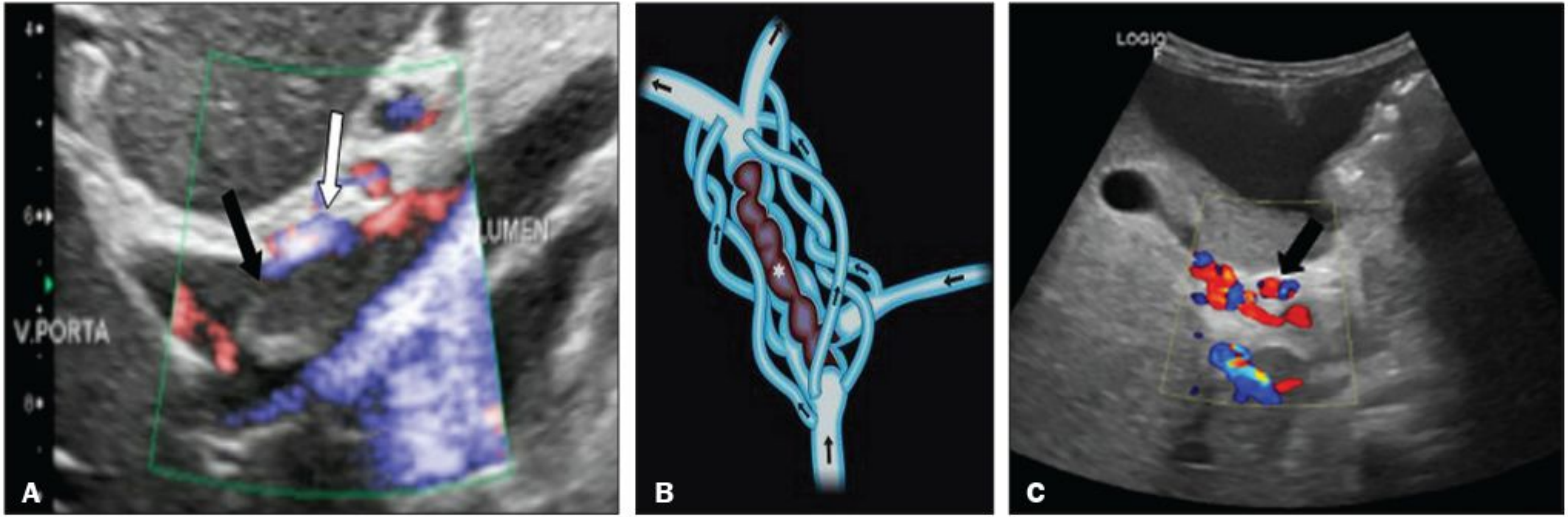
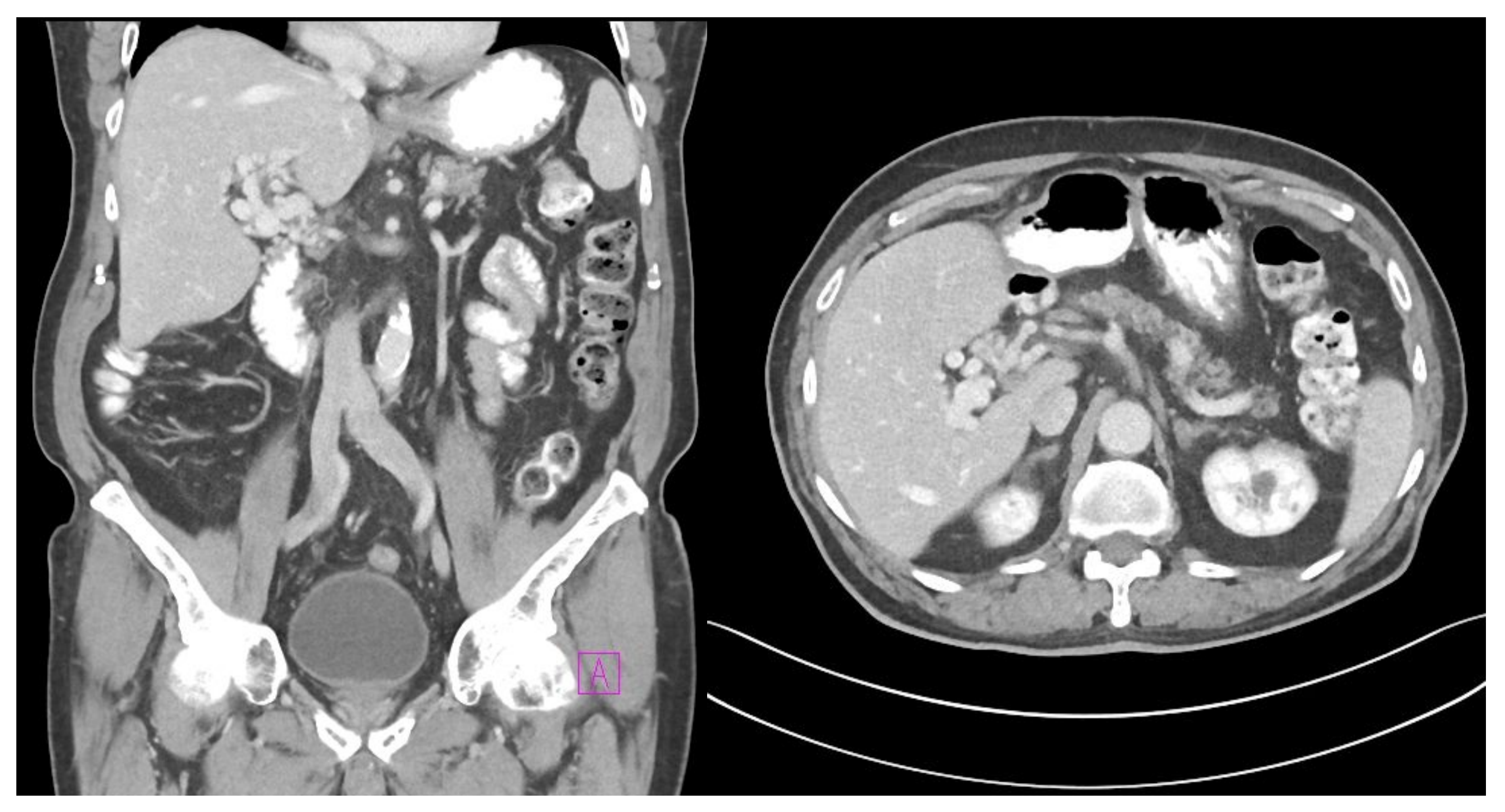
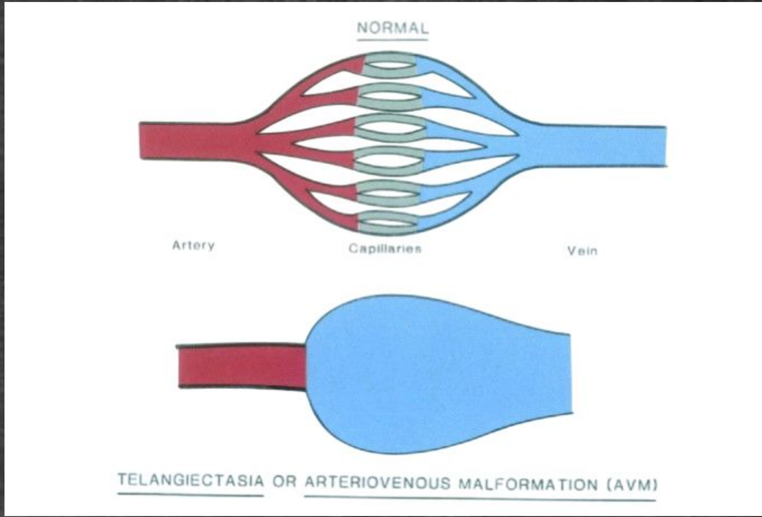


Figura 3. A: Paciente com estado de hipercoagulabilidade apresentando extensa trombose aguda da veia porta. Corte ultrassonográfico longitudinal mostrando trombo hipoecoico (seta preta) na veia porta principal, que se encontra de calibre aumentado, com discreto fluxo periférico no Doppler colorido (seta branca), que se trata de fluxo relacionado à oclusão parcial e não a fluxo no interior do trombo (o que o diferencia de um trombo tumoral). **B:** Desenho esquemático mostrando a transformação cavernomatosa da veia porta. Trombo extenso (asterisco) na veia porta principal, com a formação de múltiplas colaterais (setas) contornando a obstrução. **C:** Trombose crônica da veia porta vista na US com Doppler colorido mostrando vasos serpiginosos (seta) na região periportal, inferindo transformação cavernomatosa. **D:**


A

TELANGIECTASIA HEMORRÁGICA HEREDITÁRIA

- Síndrome Osler-Weber-Rendu;
- Autossômica dominante;
- MAV → aumenta o risco com a idade e pode envolver pele, mucosas, fígado, pulmão, TGI e SNC;
- Hemorragias, insuficiência cardíaca de alto débito por derivação hepática;
- Epistaxes recorrentes desde a infância, telangiectasias cutâneas ou mucosas, envolvimento visceral e história familiar.

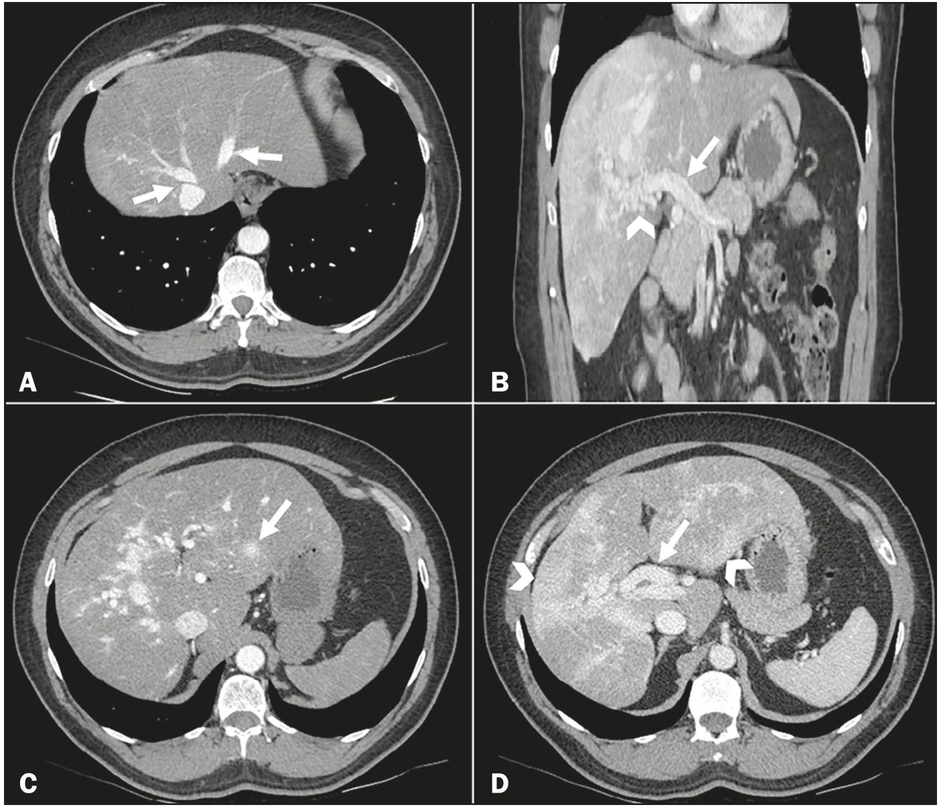


3 achados → diagnóstico

TELANGIECTASIA HEMORRÁGICA HEREDITÁRIA

- Fígado é o órgão abdominal mais acometido → durante a vida adulta;
- 8 % → fenômeno do roubo arterial
  - Shunts arterioportais ou arteriovenosos que evoluem para colangiopatia isquêmica;
  - Aumento do fluxo portal decorrente de shunts que evoluem para hipertensão portal;
  - Shunts portovenosos que resultam em insuficiência cardíaca de alto débito;
- HNF-like em 47% dos casos
  - Aparência típica
  - Falta a cicatriz central, realce arterial heterogêneo ou ausente

Figure 12. Hereditary hemorrhagic telangiectasia (HHT) in a 48-year-old woman. Coronal venous phase gadobutrol-enhanced MR image shows enlarged hepatic arteries (arrowheads), which are characteristic of liver involvement in HHT. An isoenhancing FNH-like lesion (arrow) is a common finding in these patients.



Tomografia computadorizada do abdome. A: Corte axial mostrando opacificação precoce das veias hepáticas na fase arterial (setas). B: Corte coronal demonstrando opacificação heterogênea da veia porta durante a fase portal (seta), associada a estruturas vasculares ectasiadas numerosas junto ao hilo hepático, representando uma malformação arteriovenosa (cabeça de seta). C: Corte axial identificando massa vascular confluente, medindo 1,4 cm, localizada no segmento II (seta). D: Corte axial mostrando extensa alteração perfusional do parênquima hepático, caracterizada como padrão em mosaico (cabeças de setas), além de artéria hepática de calibre aumentado (seta)

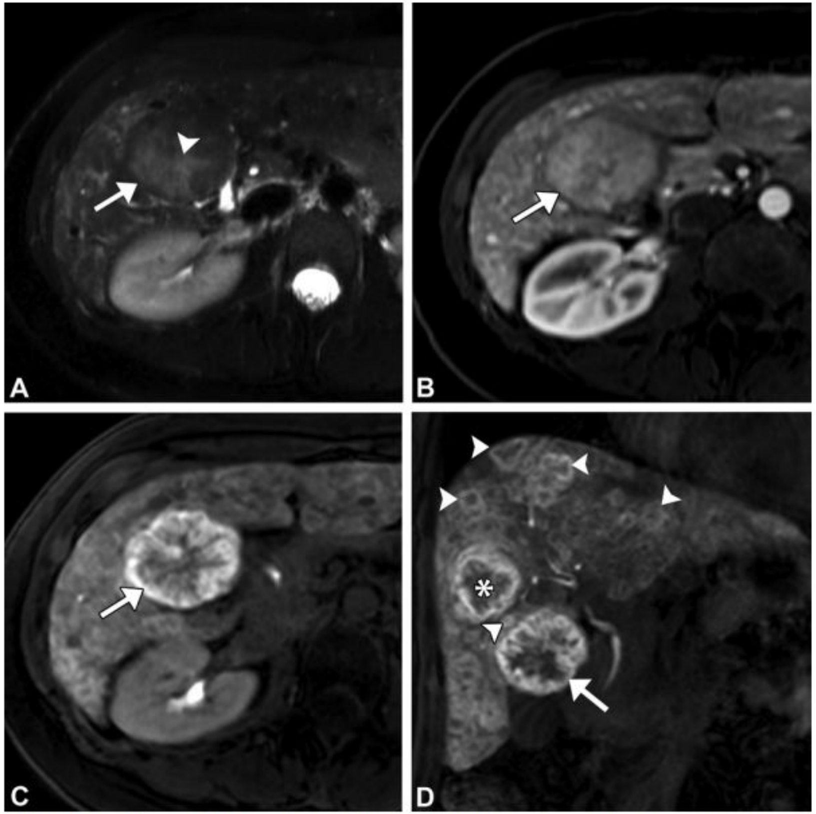


Figure 13. HHT and FNH-like lesions (arrow) with typical features at MRI in a 44-year-old woman. **(A)** Axial precontrast T2-weighted fat-saturated MR image shows characteristic features of an FNH-like lesion, with a T2-hyperintense central scar (arrowhead). **(B)** Axial gadoxetate-enhanced arterial phase MR image shows homogeneous APHE. **(C, D)** Axial **(C)** and coronal **(D)** gadoxetate-enhanced HBP MR images show homogeneous hyperintense uptake. The coronal image **(D)** shows multiplicity, with an additional FNH-like lesion (*) and numerous smaller FNH-like lesions (arrowheads) that were indiscernible from the background of telangiectases and "large confluent vascular masses" that were described in reports from a prior MRI examination (not shown) performed with an extracellular contrast agent.

Table 1: Conditions Predisposing Patients to Formation of FNH-like Lesions

Abnormalities of hepatic outflow
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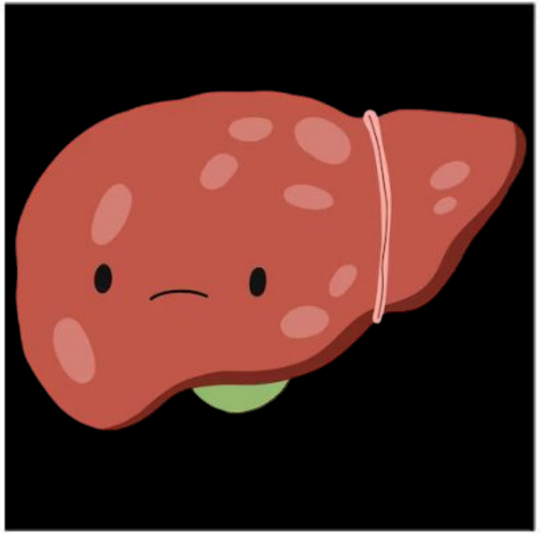
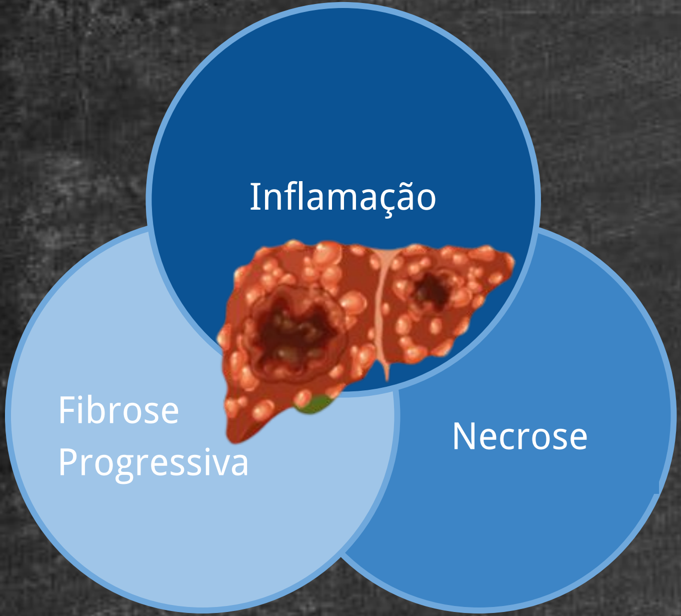
Cavernous transformation
Hereditary hemorrhagic telangiectasia
Hepatic microvascular disturbances
Cirrhosis
Nodular regenerative hyperplasia
Chemotherapy induced

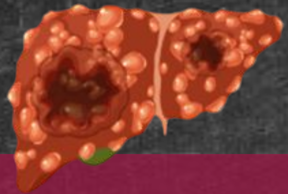


HNF-like

CIRROSE



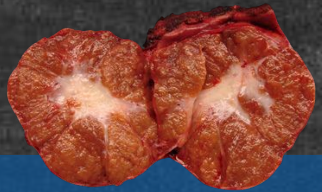
- HNF-like → 3-15%;
  - Dinâmica vascular alterada:
    - Remodelamento sinusoidal;
    - Isquemia e angiogênese local;
    - Shunts intra-hepáticos;
  - Don’t touch lesions;



CHC

- Realce arterial;
- *Washout;*
- Pseudocápsula;
- HBP → hipointenso
  - 6-15% → iso ou hiperintenso.

- T2 → hiperintenso;
- DWI → aumento de sinal;
- *Washout* → S: 90% e E: 100%



HNF-like

- Realce arterial;
- *Washout;*
- Pseudocápsula;

$\left.\right\}$ 6-10%

- HBP → iso ou hiperintenso;
  - 30% → hipointenso.

- Cicatriz central;
- < 1,6 cm → DWI: iso ou hipo;
- Ausência de *washout* → E: 100%

LI-RADS Category and Interpretation

LR-1	Definitely benign (100% certainty)
LR-2	Likely benign
LR-3	Intermediate likelihood of HCC
LR-4	Likely HCC
LR-5	Definitely HCC (100% certainty)
LR-TIV	Definite tumor in vein (not specific for HCC)
LR-M	Likely malignant (not specific for HCC)

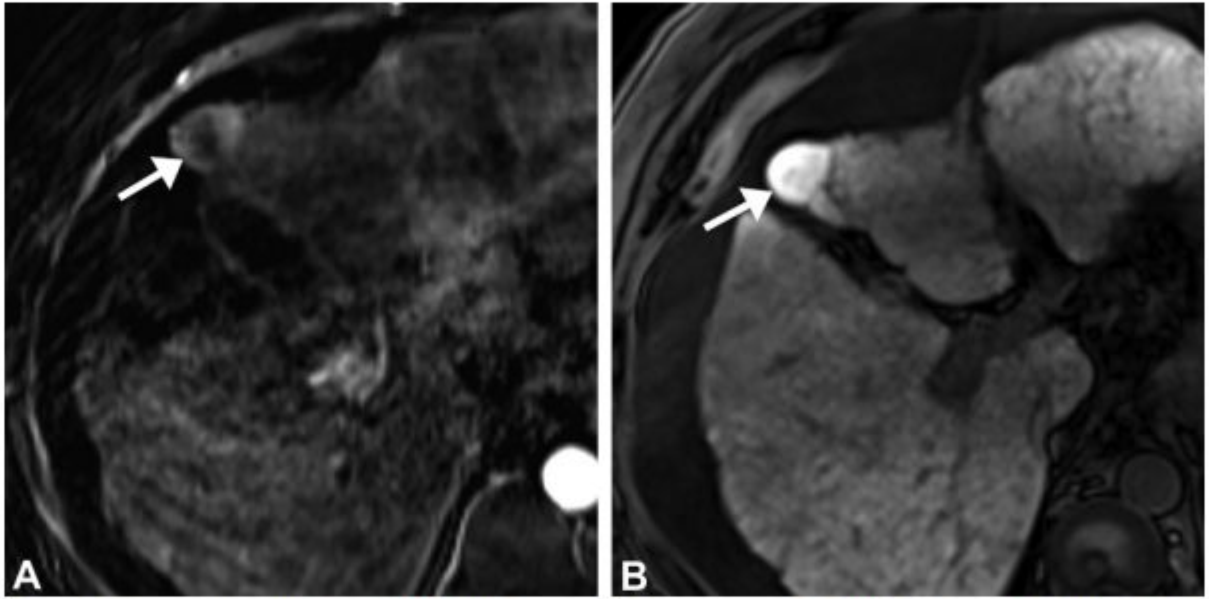


Figure 14. Pathologically proven FNH-like lesion in a 56-year-old man with cirrhosis. **(A)** Axial gadoxetate-enhanced arterial phase subtraction MR image shows APHE (arrow). **(B)** Axial gadoxetate-enhanced HBP MR image shows typical homogeneous hyperintense contrast material uptake (arrow).

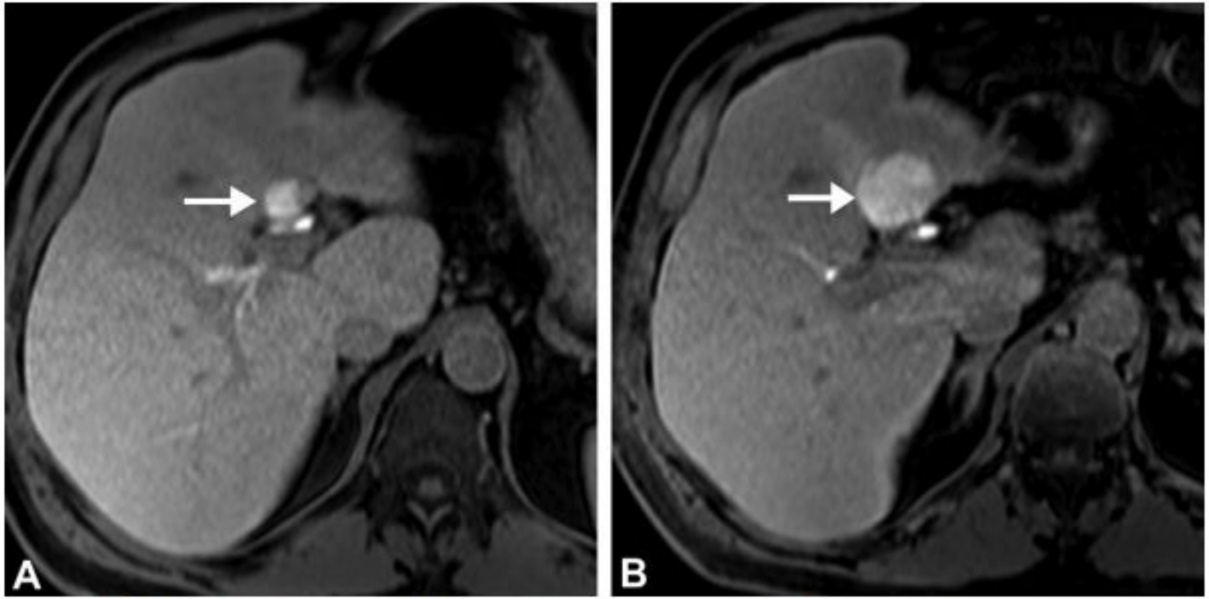


Figure 15. Pathologically proven HCC in a 68-year-old man with cirrhosis. **(A)** Axial gadoxetate-enhanced HBP MR image shows the lesion (arrow) with homogeneous hyperintense uptake at baseline. **(B)** Axial gadoxetate-enhanced HBP MR image at 6-month follow-up shows growth of the lesion (arrow), suggesting the correct diagnosis of HCC.

HIPERPLASIA NODULAR REGENERATIVA

- Alteração vascular rara;
- Nódulos regenerativos difusos (< 3 mm);
- Mínima fibrose;
- Principal causa intra-hepática de hipertensão portal não cirrótica;
- Insultos microvasculares → atrofia do parênquima → isquemia hepatocitária → regeneração nodular;
- Maioria assintomática.

Table 3: Conditions Associated with NRH

Hematologic
Lymphoproliferative disorders
Myeloproliferative disorders
Common variable immunodeficiency syndrome
HIV infection
Autoimmune
Inflammatory bowel disease
Rheumatoid arthritis
Systemic lupus erythematosus
Vascular
Congenital absence of the portal vein
Portal vein thrombosis
Budd-Chiari syndrome
Medications
Chemotherapeutics
Immunosuppressants
Antivirals

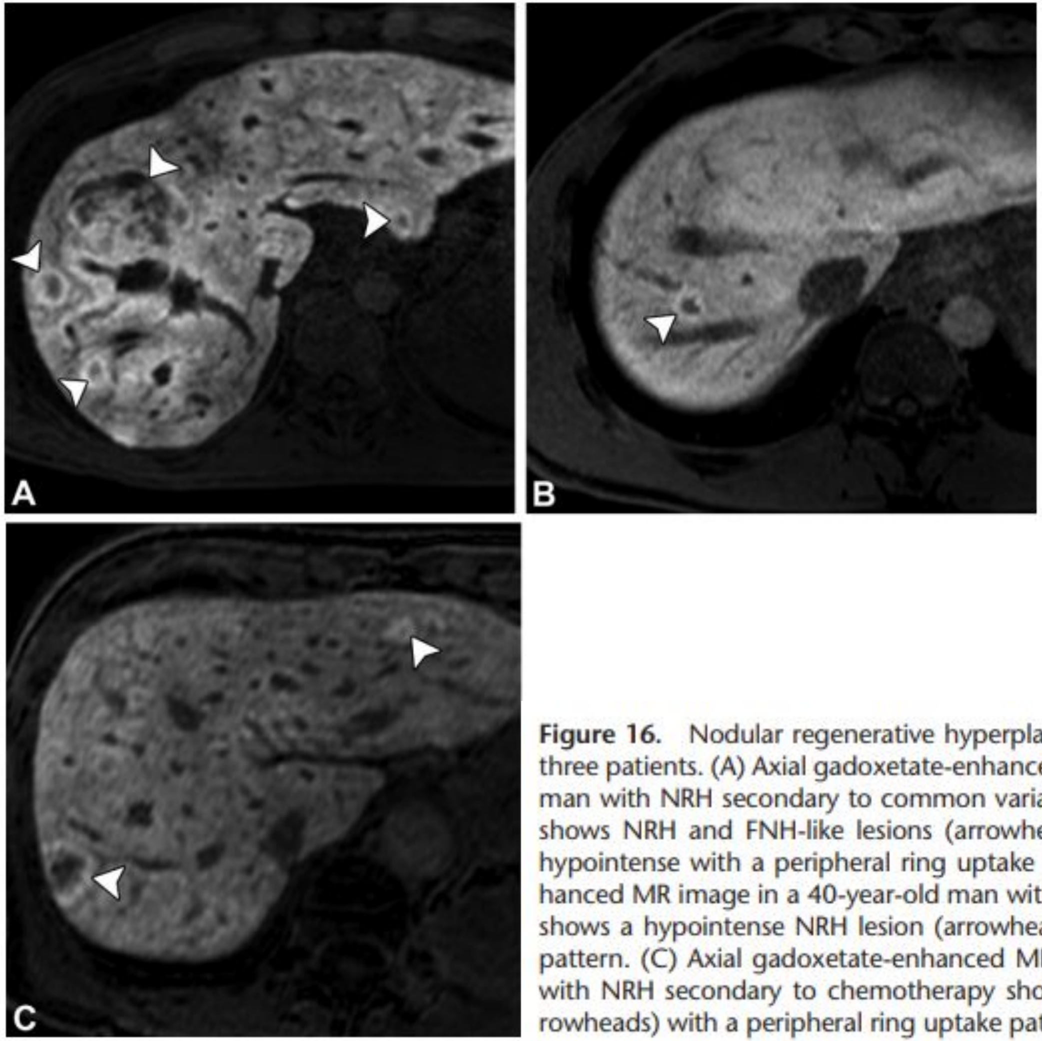
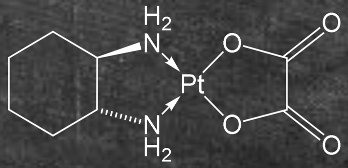
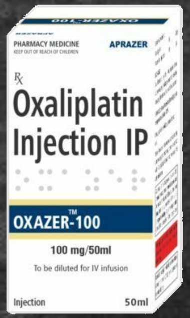


Figure 16. Nodular regenerative hyperplasia (NRH) and FNH-like lesions in three patients. (A) Axial gadoxetate-enhanced HBP MR image in a 44-year-old man with NRH secondary to common variable immunodeficiency syndrome shows NRH and FNH-like lesions (arrowheads) that appear predominantly hypointense with a peripheral ring uptake pattern. (B) Axial gadoxetate-enhanced MR image in a 40-year-old man with NRH secondary to HIV infection shows a hypointense NRH lesion (arrowhead) with a peripheral ring uptake pattern. (C) Axial gadoxetate-enhanced MR image in a 65-year-old woman with NRH secondary to chemotherapy shows hypointense NRH lesions (arrowheads) with a peripheral ring uptake pattern.

QUIMIOTERAPIA

- Injúria hepática aguda, esteatose, esteato-hepatite, síndrome de obstrução sinusoidal, HNF-like;
- Achados típicos;
- Diferenciar de metástases





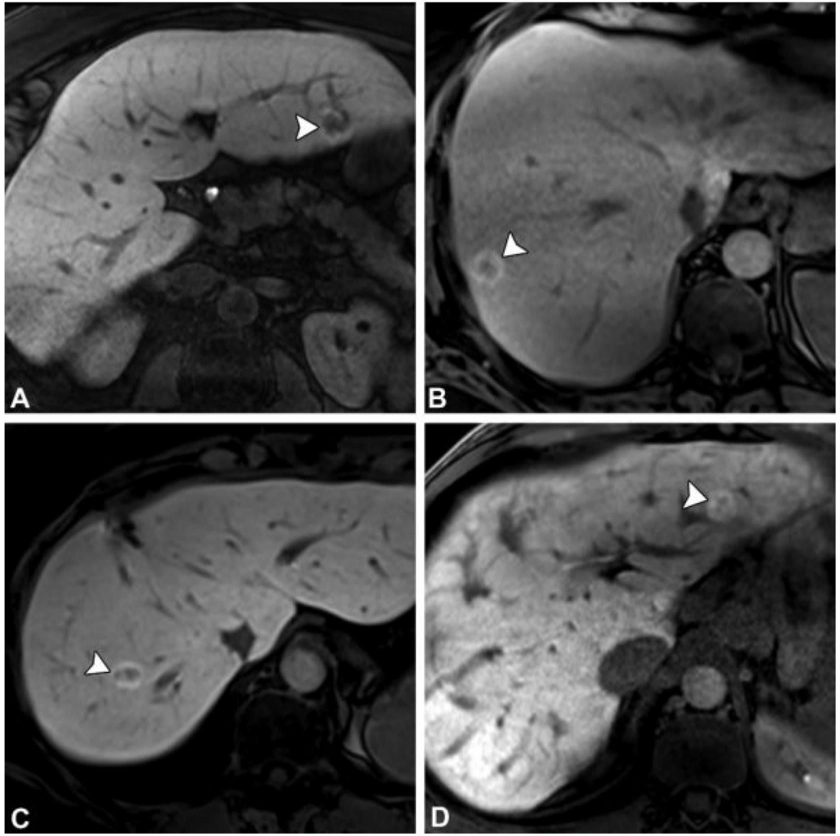


Figure 17. New FNH-like lesions in four patients after treatment with oxaliplatin-based chemotherapy. **(A–C)** Axial gadoxetate-enhanced HBP MR images in a 60-year-old woman who was treated for colon adenocarcinoma **(A)**, a 49-year-old man who was treated for rectal adenocarcinoma **(B)**, and a 62-year-old woman who was treated for colon adenocarcinoma **(C)** show a hypointense FNH-like lesion (arrowhead) with a peripheral rim uptake pattern. **(D)** Axial gadoxetate-enhanced HBP MR image in a 38-year-old man who was treated for colon adenocarcinoma shows an FNH-like lesion (arrowhead) with a homogeneously hyperintense uptake pattern.

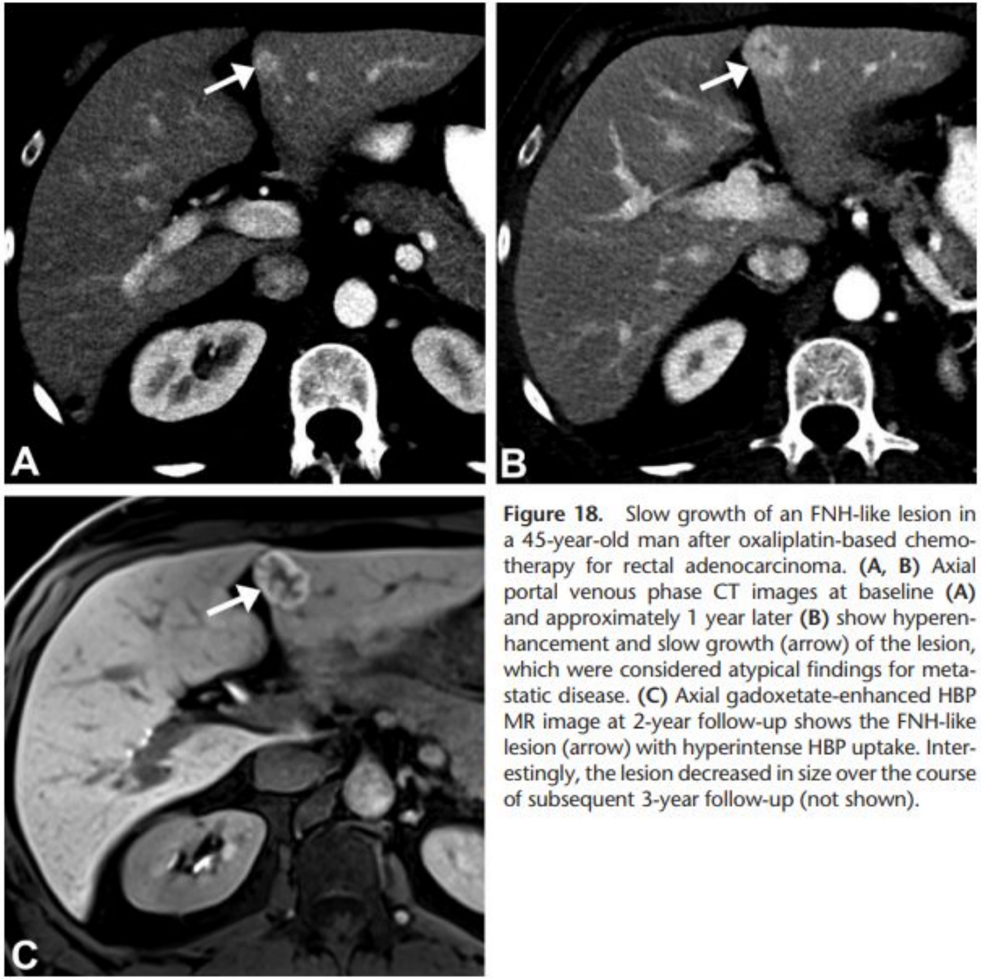


Figure 18. Slow growth of an FNH-like lesion in a 45-year-old man after oxaliplatin-based chemotherapy for rectal adenocarcinoma. **(A, B)** Axial portal venous phase CT images at baseline **(A)** and approximately 1 year later **(B)** show hyperenhancement and slow growth (arrow) of the lesion, which were considered atypical findings for metastatic disease. **(C)** Axial gadoxetate-enhanced HBP MR image at 2-year follow-up shows the FNH-like lesion (arrow) with hyperintense HBP uptake. Interestingly, the lesion decreased in size over the course of subsequent 3-year follow-up (not shown).

OBRIGADA!